CCH® Tagetik | **ACADEMY**

Planning and forecasting advanced



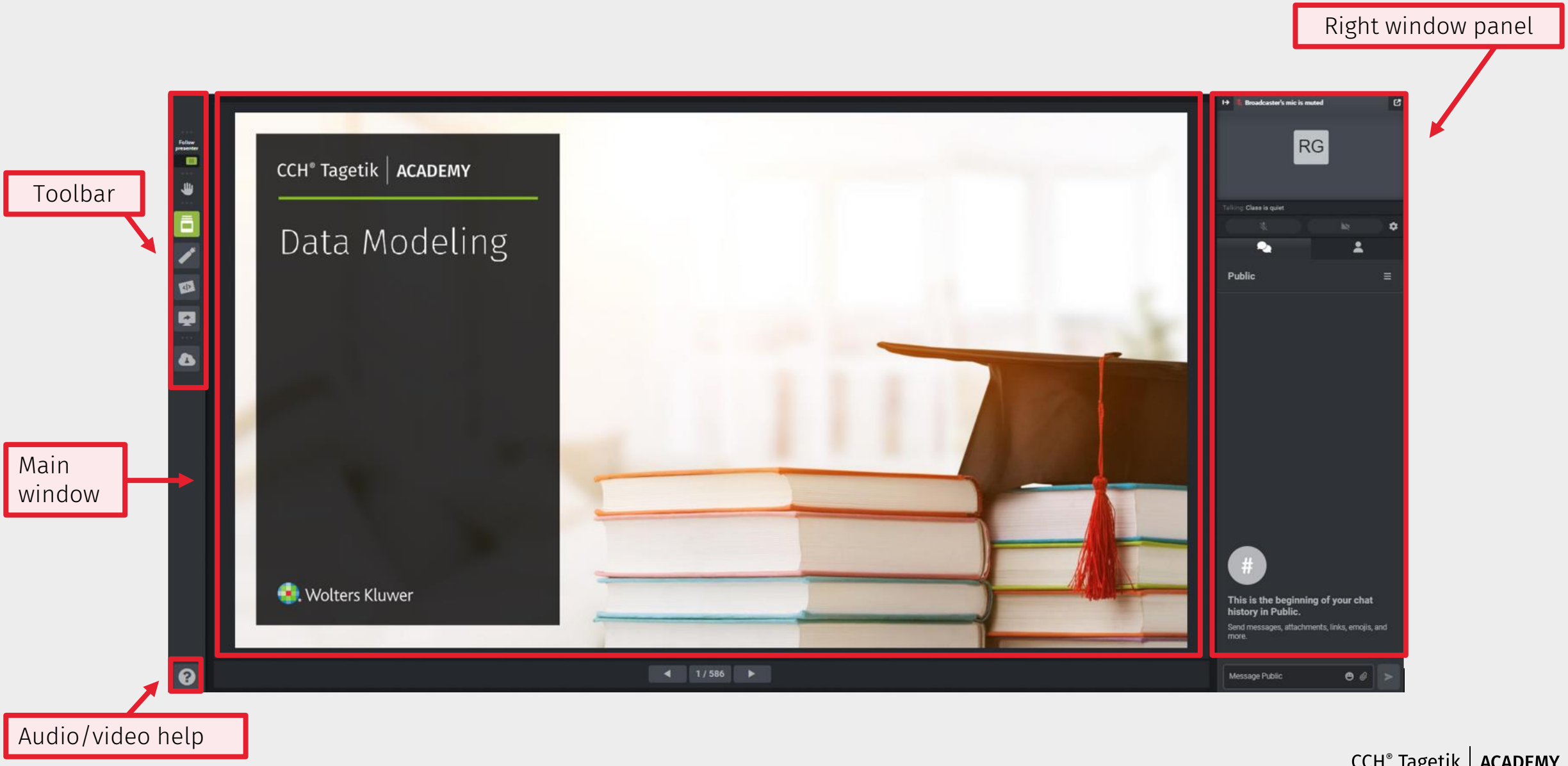
Wolters Kluwer

Before starting

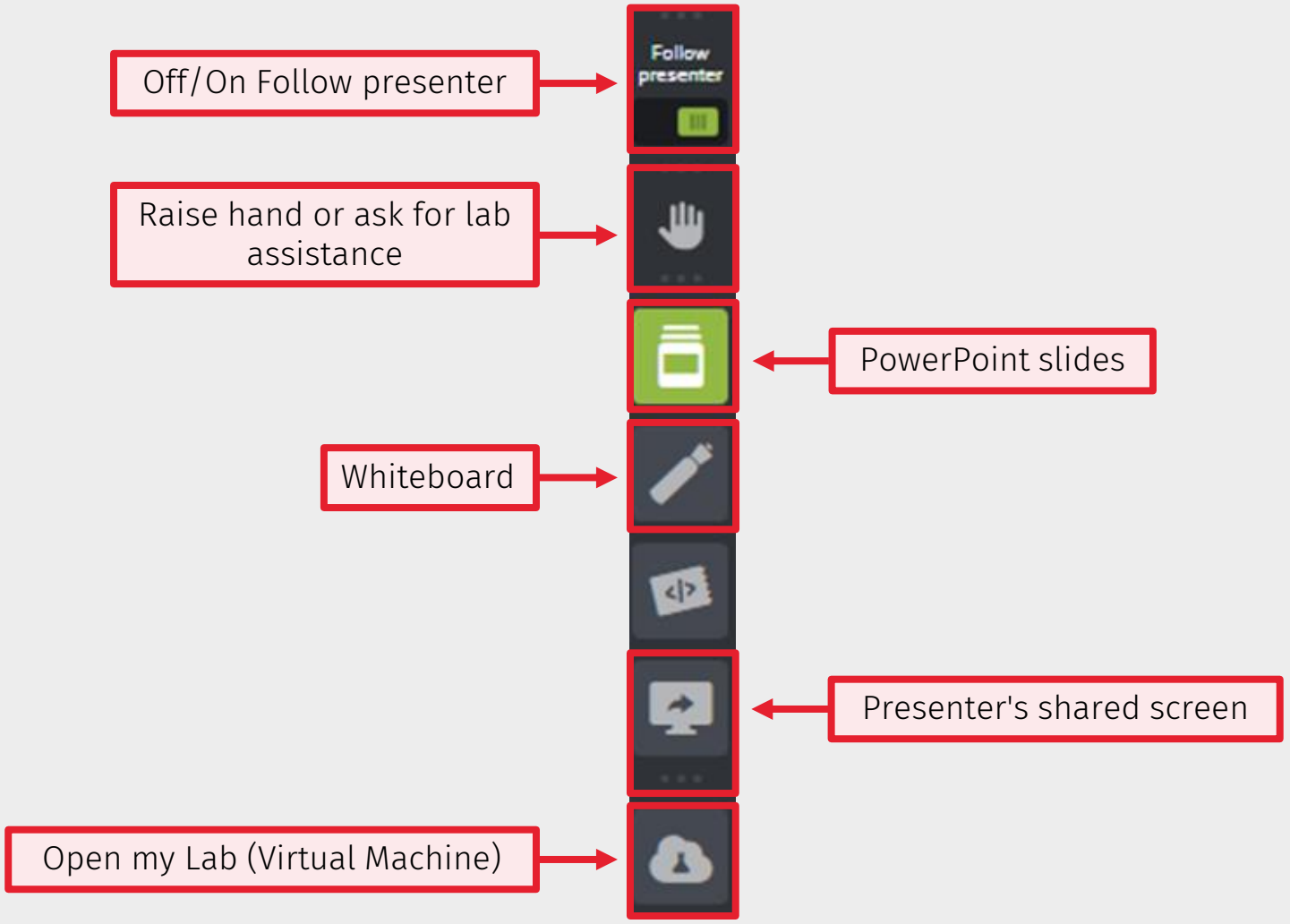
Strigo platform - “How to”



Strigo overview (for Instructor-led classes)



Strigo left toolbar

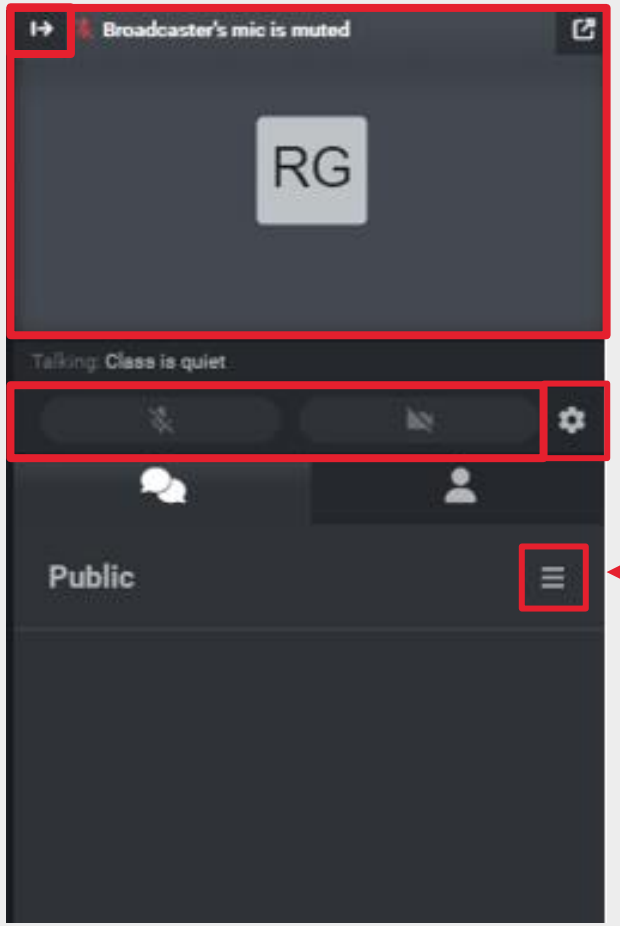


Strigo right window panel

Hide/show window panel

Web cam presenter & who's talking

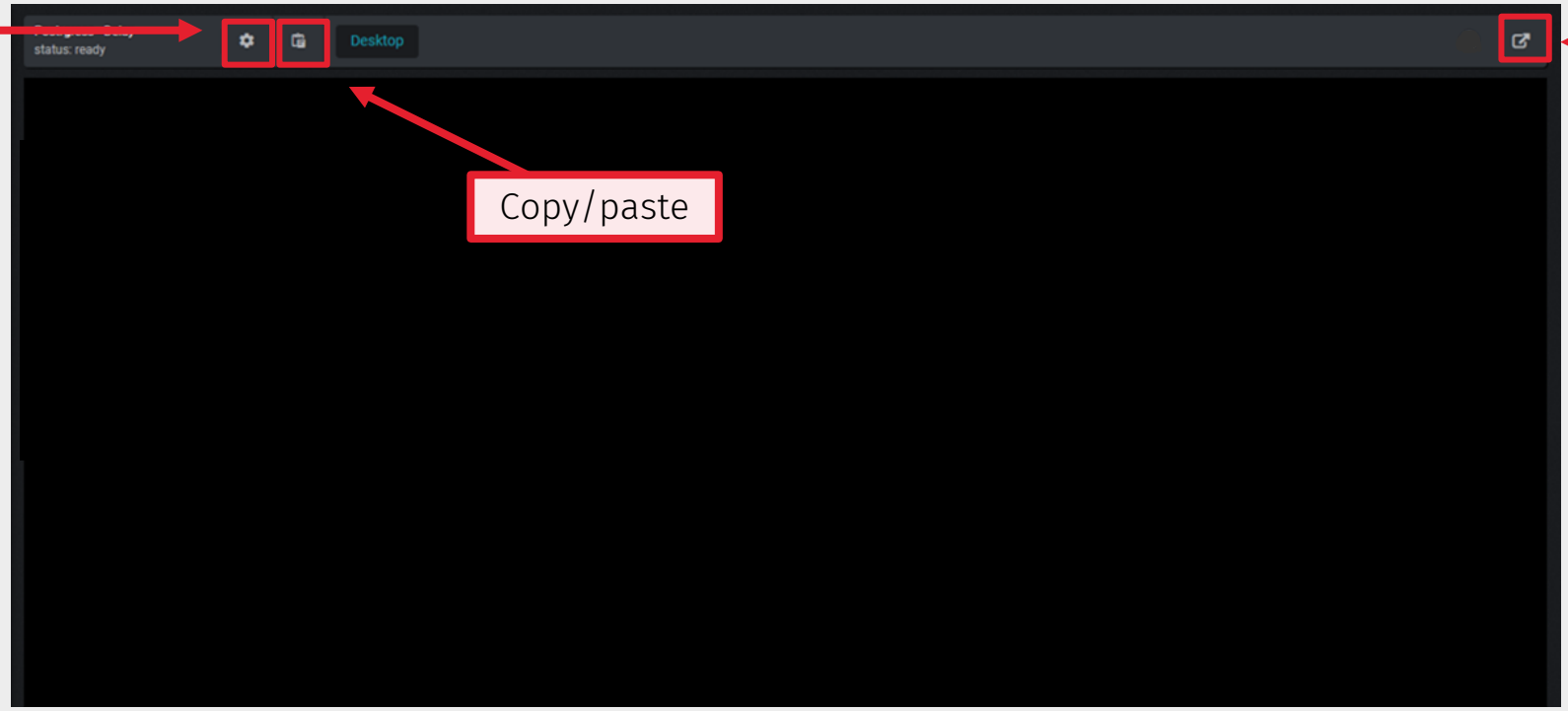
Off/On microphone and camera



A/V Troubleshoot

Open chat menu to switch between public chat and private chat

Strigo Lab (Virtual Machine)



Agenda



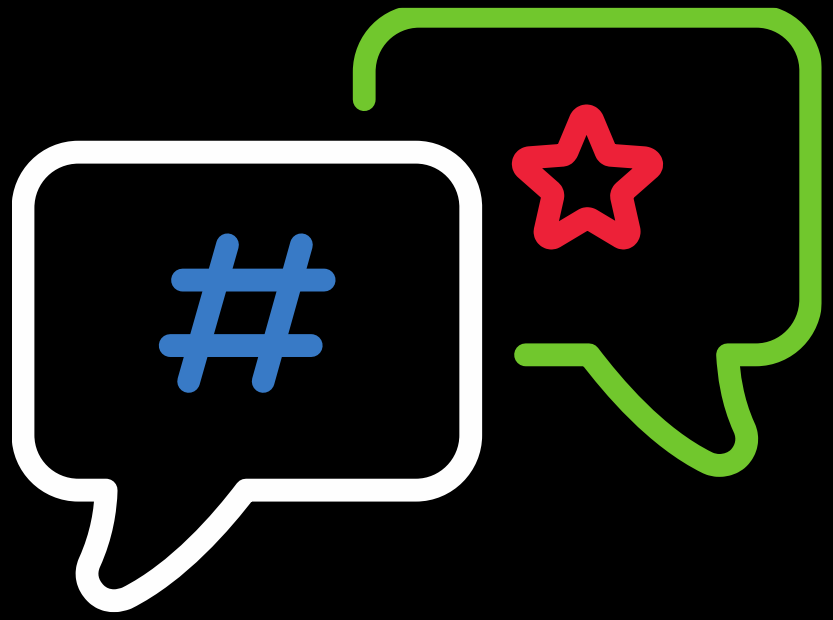
- Solution architecture
- Loading tools
- DTP performances
- Time frame management
- Comparisons with other data
- Global assumption management
- DTP calculations
- Other calculation tools
- Allocation logics (optional)

Business cases



- Budget population and update of data model
- Multi-year planning by projects
- Comparison with actual and budget adjustments
- Forecast

Let’s talk about: Solution architecture



Planning on CCH Tagetik

Analytic Information Hub

Allows you to manage different data models suitable for any planning process function.

Predictive intelligence

Provides estimates based on machine learning.

Financial workspace

Provides built-in tools for financial planning.

How to approach the design

Customer requirements

- Project summary
- Specific requirements
- Workflow
- Approval hierarchies
- Dimensions

Customer files

- Data dump
- DB structure
- List of elements
  - Cost centers
  - Entities
  - Region
  - Etc.

Workshops

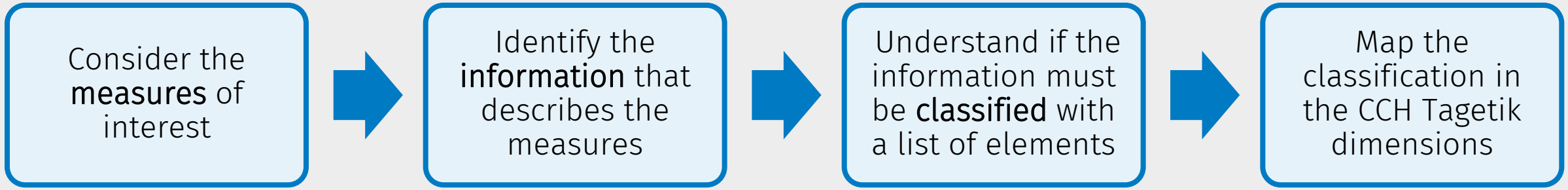
- Additional information
- Context
- Q&A

Solution design template

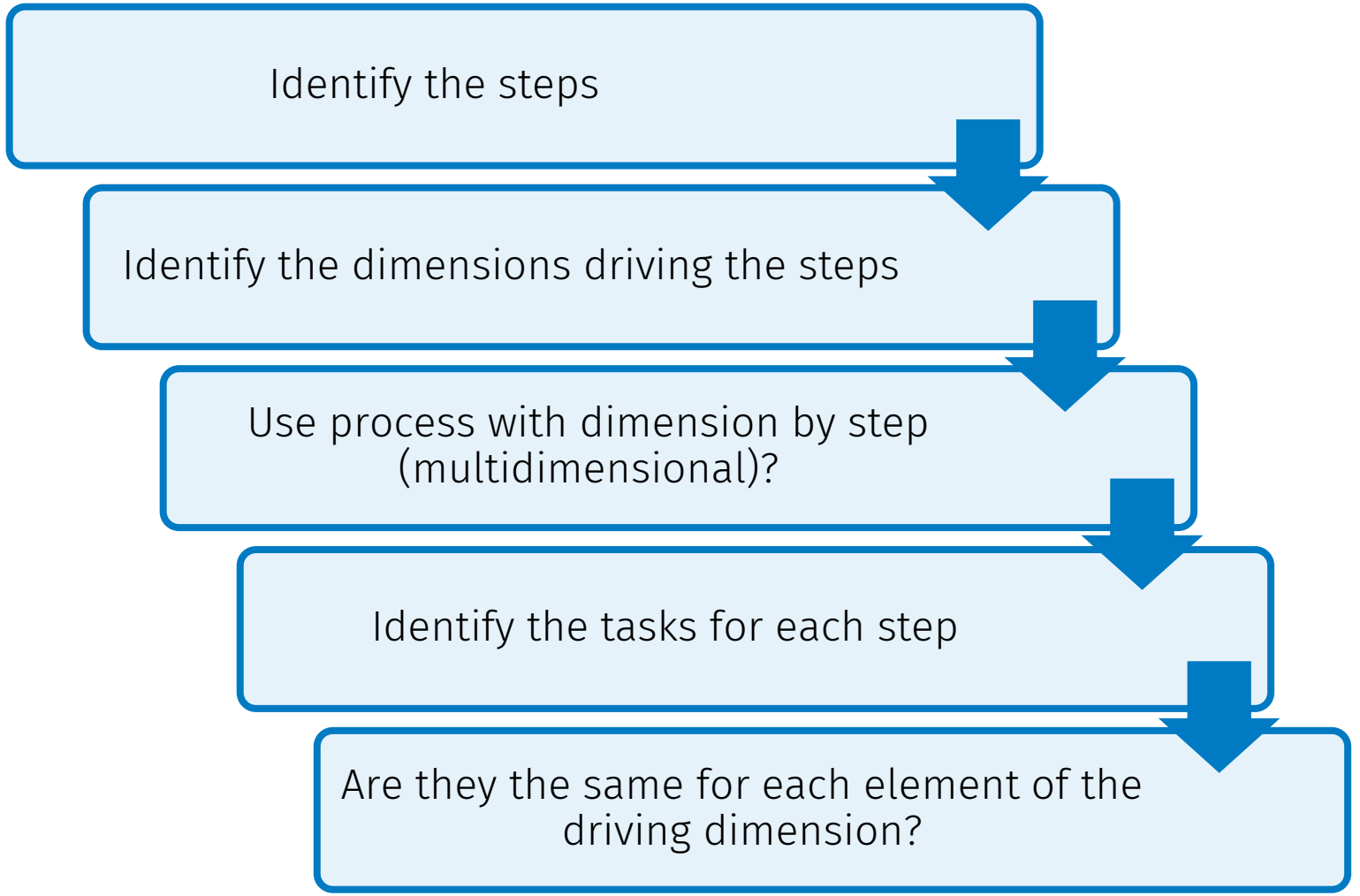
To be filled out with:

- Application blueprint
- Data model
- Workflow
- Approval process

Dimension design



Process design



Dataset design

Analyze data structure and data processing in the context of the workflow to identify:

- Datasets
- How data moves from a table to another

Define the information within the single dataset to find the suitable structure, also considering technical aspect

- Alternative structure for the same information (e.g. periods in different columns)
- Specific fields for calculation
- Peculiar configuration (e.g. dimensions handled as texts)
- Definition of the dataset as reference datasets
- Etc.

Data integration, data processing and built-in tools: questions

How is data collected and what are the necessary transformation to integrate it in the data model implemented?

How is data processed within CCH Tagetik?

Is it possible to leverage built-in functionalities, predefined configurations or common best practices?

What are the most suitable tools to use?

Data integration, data processing and built-in tools: suggestions

Verify a wide range of alternative solutions, their pros and cons and quick prototyping to test their validity

Test if it is possible to enhance performance

Planning vs other processes

Design should take into consideration that

- There might be other processes to manage in the application
- Data of different processes must often be comparable

Use of dimensions

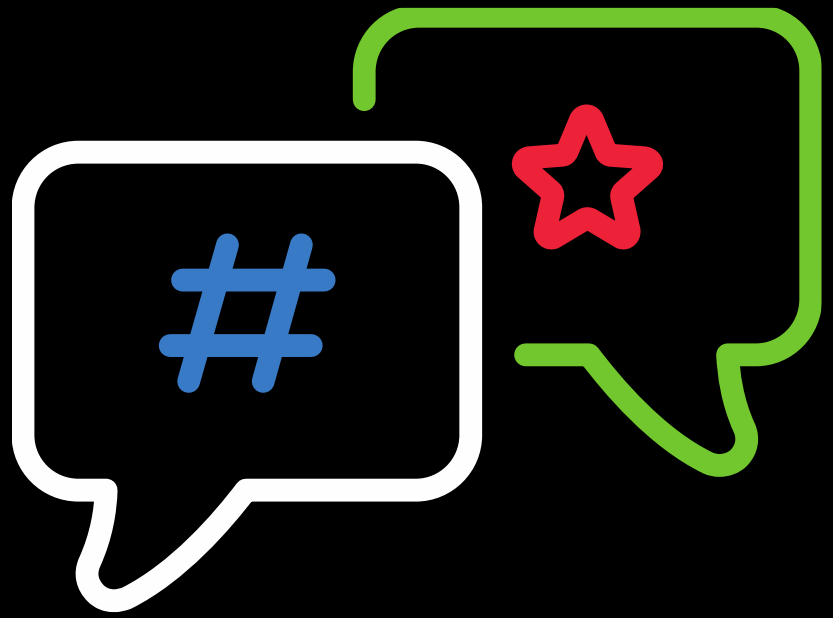
Custom dimensions classify financial data

- What is the maximum level of financial detail?

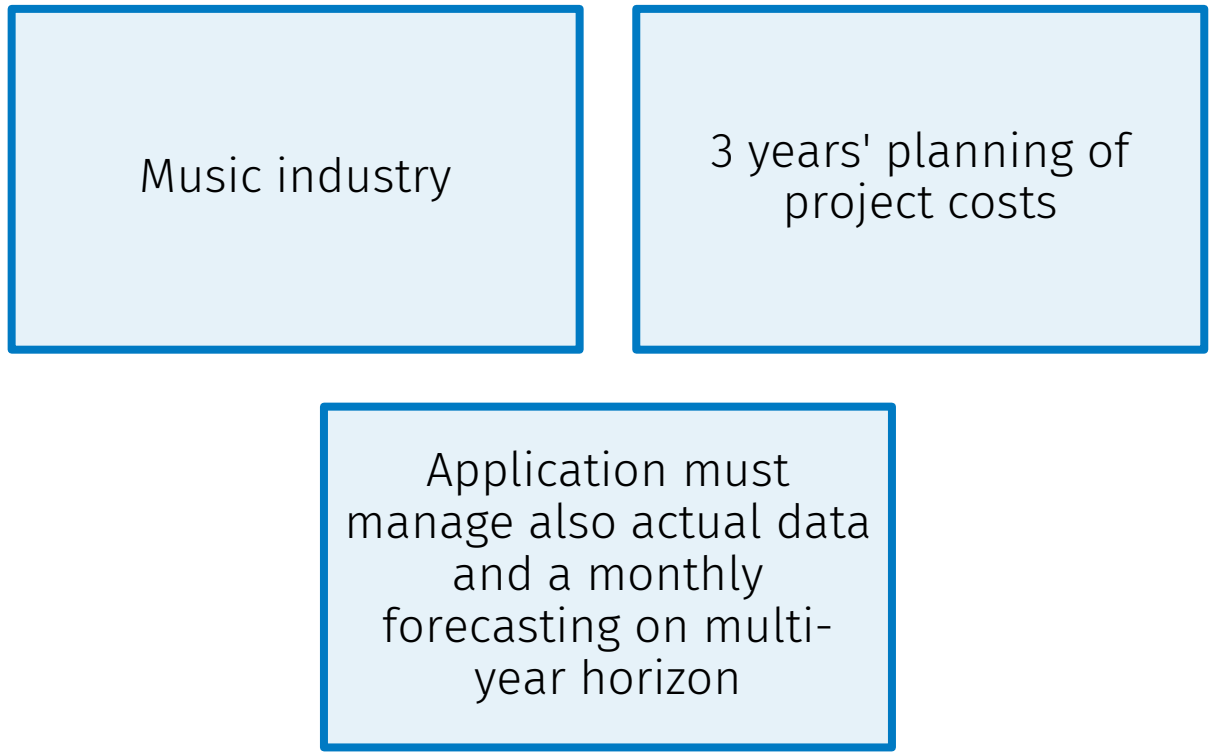
In AIH, standard dimensions

- May reduce freedom and flexibility
- Increase control and leverage built in functionalities

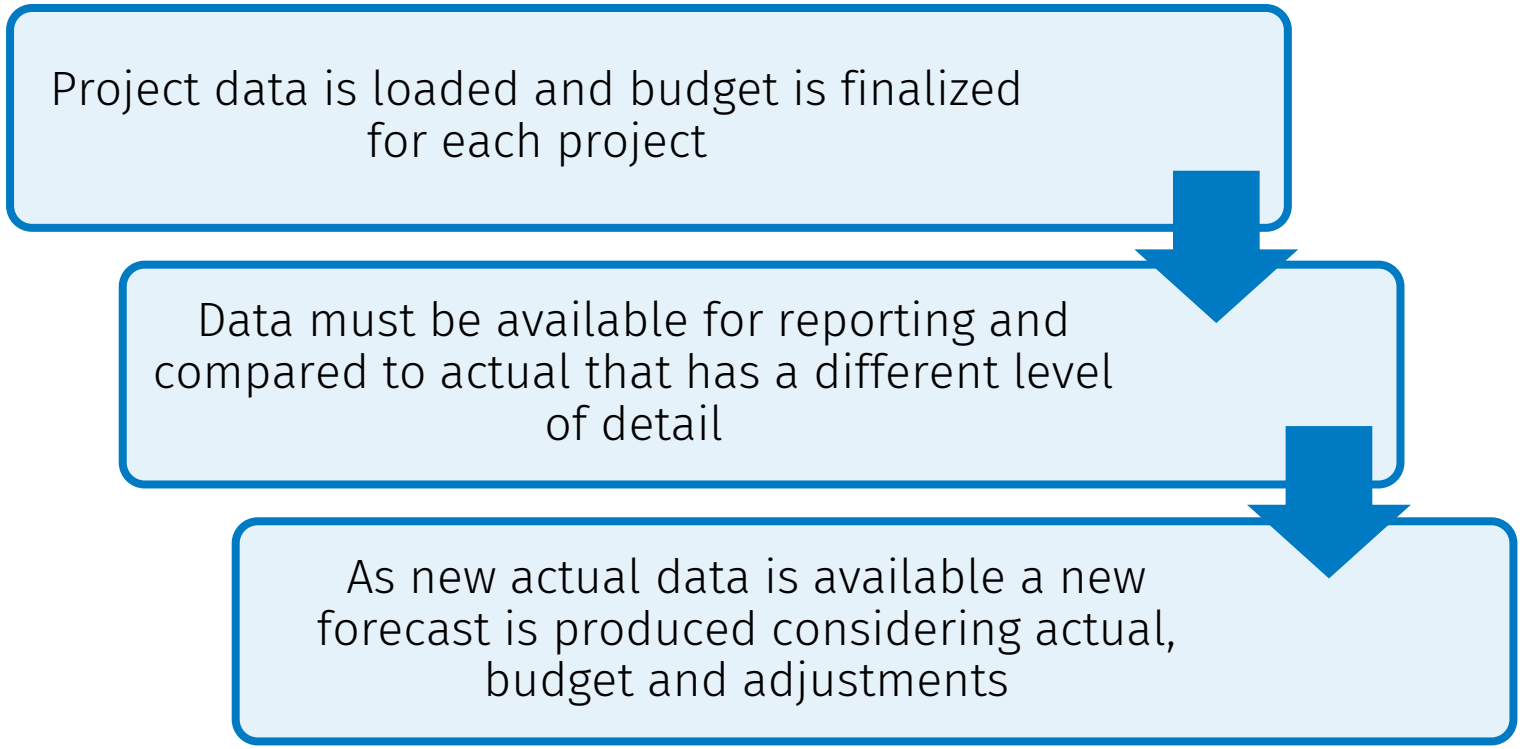
Let’s talk about: Business case – Design



Business Case Overview



Business Case Steps

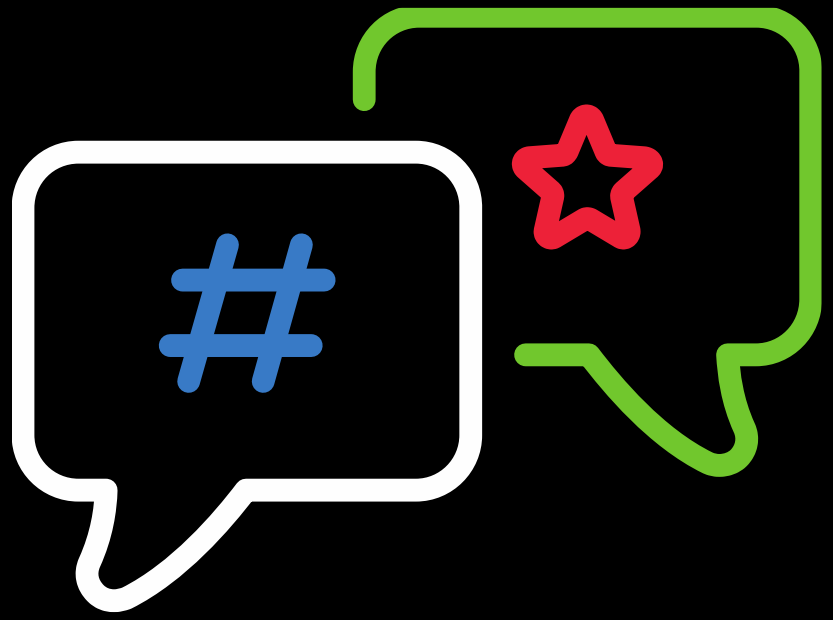


Now it’s your turn

Exercise # 1



Let’s talk about: Loading tools



Many options, not always good news



ETL

PROS

- Flexible on sources and target
- Volumes and controlled data
- Fast development
- Out of the box functionalities

CONS

- SQL skills required
- Requires knowledge of CCH Tagetik Data model
- Not tailored for AW
- Slower compared to DTP

Quick Data Loader

PROS

- Flexible on sources and target on any fact table
- Load high volumes of data on AW and FW
- Quick and easy to setup

CONS

- No data transformations
- No scheduling option
- End user handling only

Data Source mapping

PROS

- Flexible on sources
- Load high volumes of data on target dataset
- Quick and easy to setup
- Mapping defined within dataset structure
- Schedulable
- Can be run into DTP with a specific operation

CONS

- No data transformations

Datasource

PROS	CONS
• Definition based on a SQL Query that can read and join any visible table • Grouping , sum and any SQL operation available • Parameters available • Data retrieved depends on moment of access	• SQL skills required

DTP

PROS

- Tailored to create data processing on AW
- Provides many operations to create calculation
- Possibility to debug
- Fast development of staging area
- Two running modalities
- Metadata loading available

CONS

- Limited checks on data
- Strictly coupled to relational database logics
- Lack of loading filters

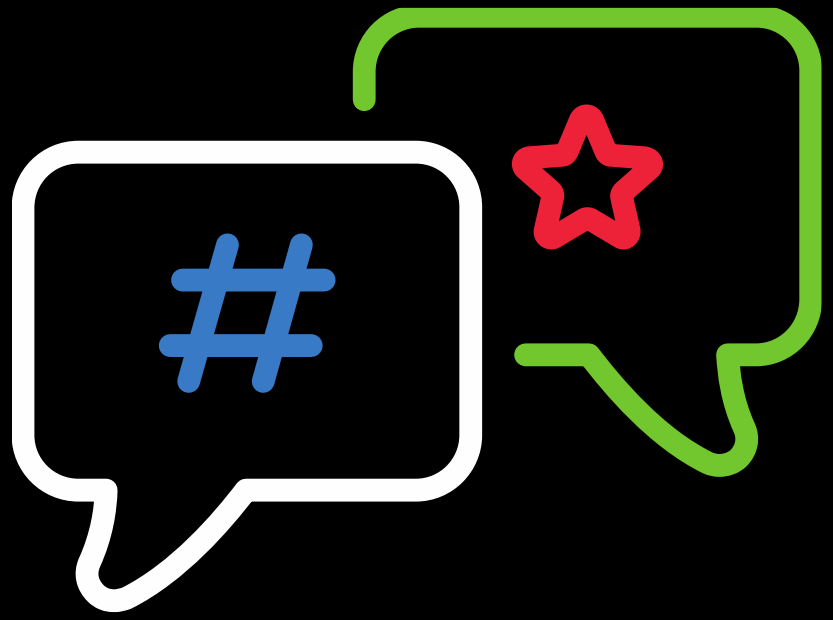
Data entry form

PROS	CONS
• Excel-based structure • Used to load adjustments and small volumes of data • Calculation with immediate feedback	• Not suitable for massive Loading/calculation • Processes data present in the form

Stored procedure

PROS	CONS
• Maximum flexibility • Fast performance	• Not a CCH Tagetik tool • Advanced SQL skills required • Dependent on implementer capability • Hard to maintain • Even harder to hand over (i.e. to support or a colleague) • Unpredictable after system update

Let’s talk about: Use of REST API in CCH Tagetik



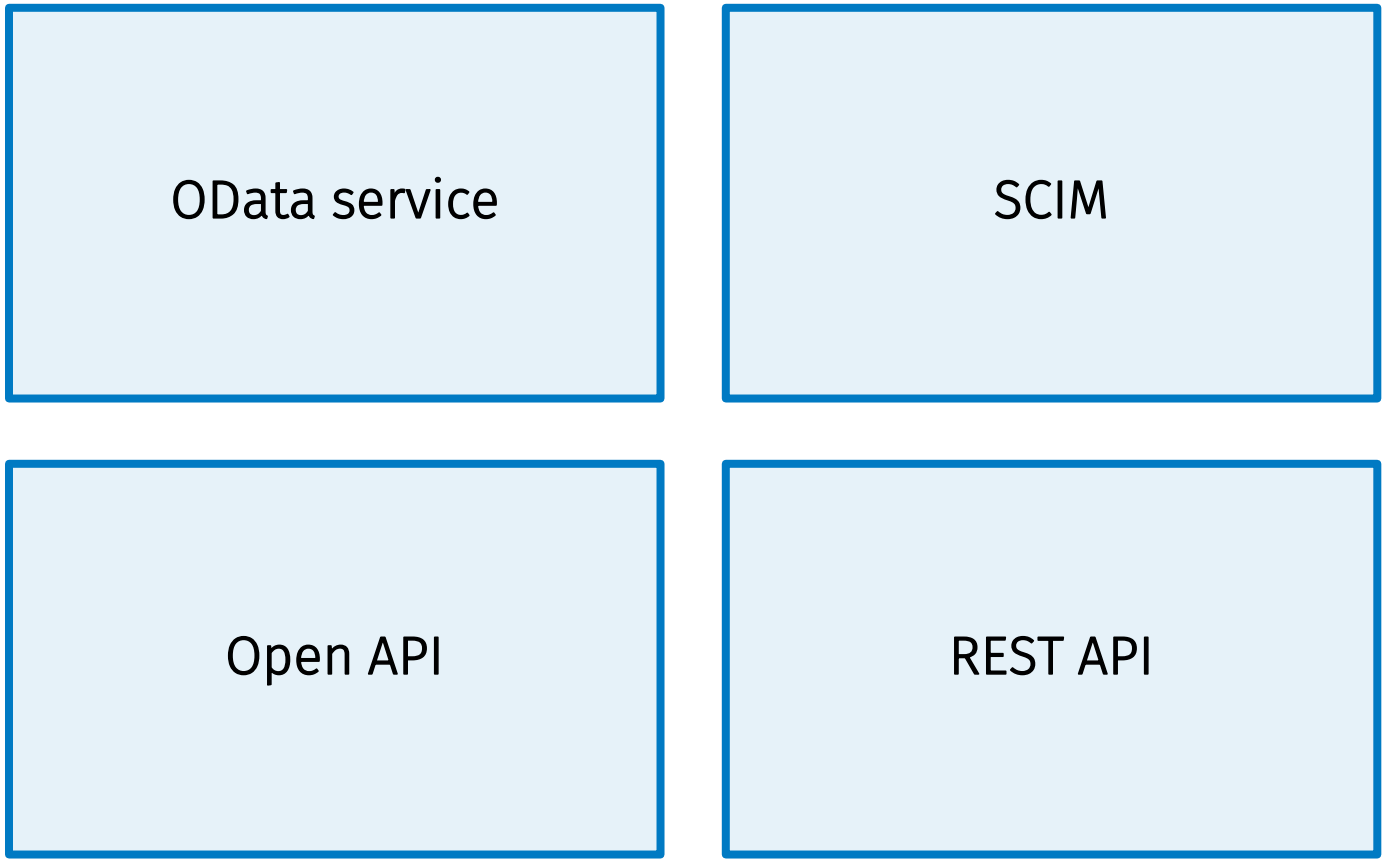
Use of REST API in CCH Tagetik

Why to use RESTful API

- Retrieve difficult to access data and metadata
- Expose AW Data toward to external applications
- Drive Data processing without accessing the CCH Tagetik application directly

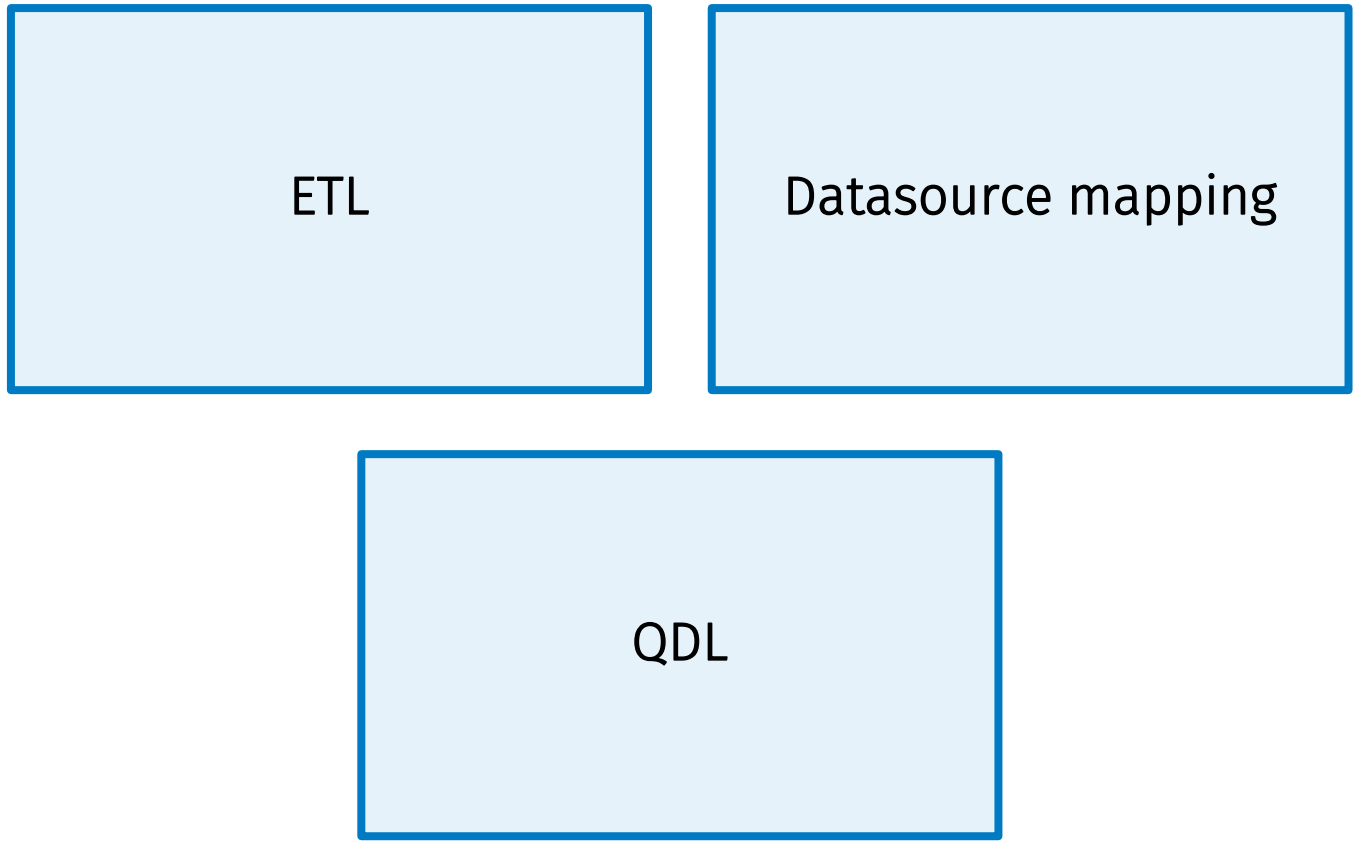
Use of REST API in CCH Tagetik

Available **standards** to implement REST API:

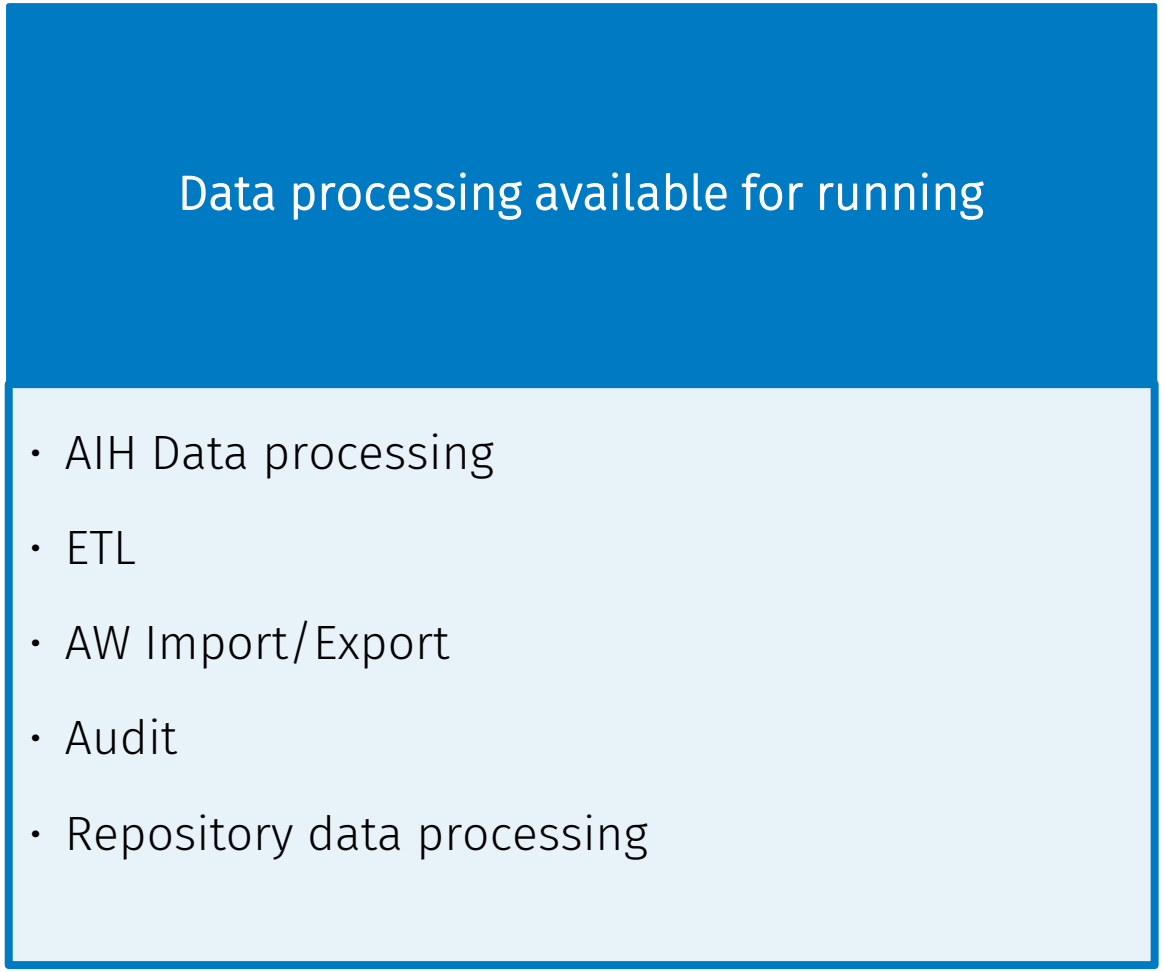


Data processing and tools driven by RESTful API

Tools to consume RESTful API



Data processing and tools driven by RESTful API



RESTful API

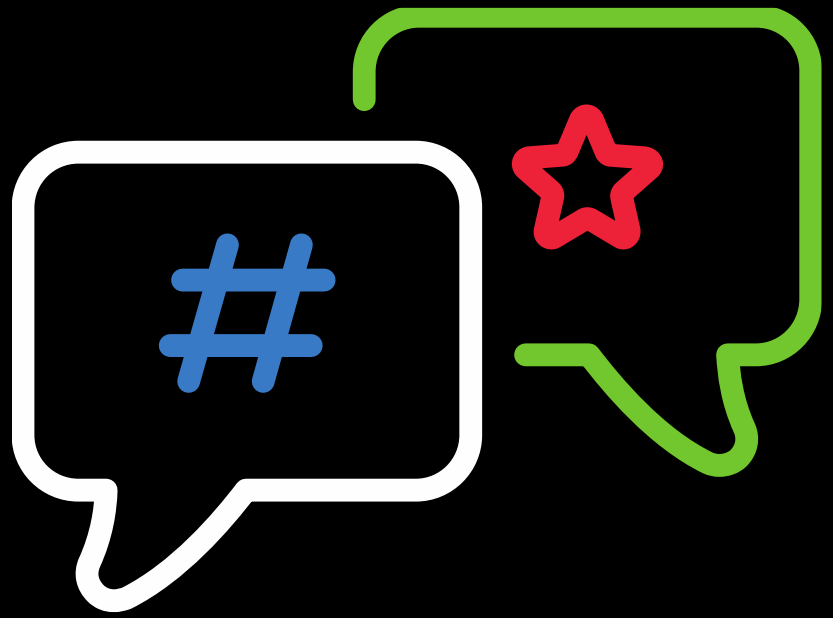
PROS

- Eases data access directly via HTTPs
- Allows data processing run without accessing the CCH Tagetik application directly

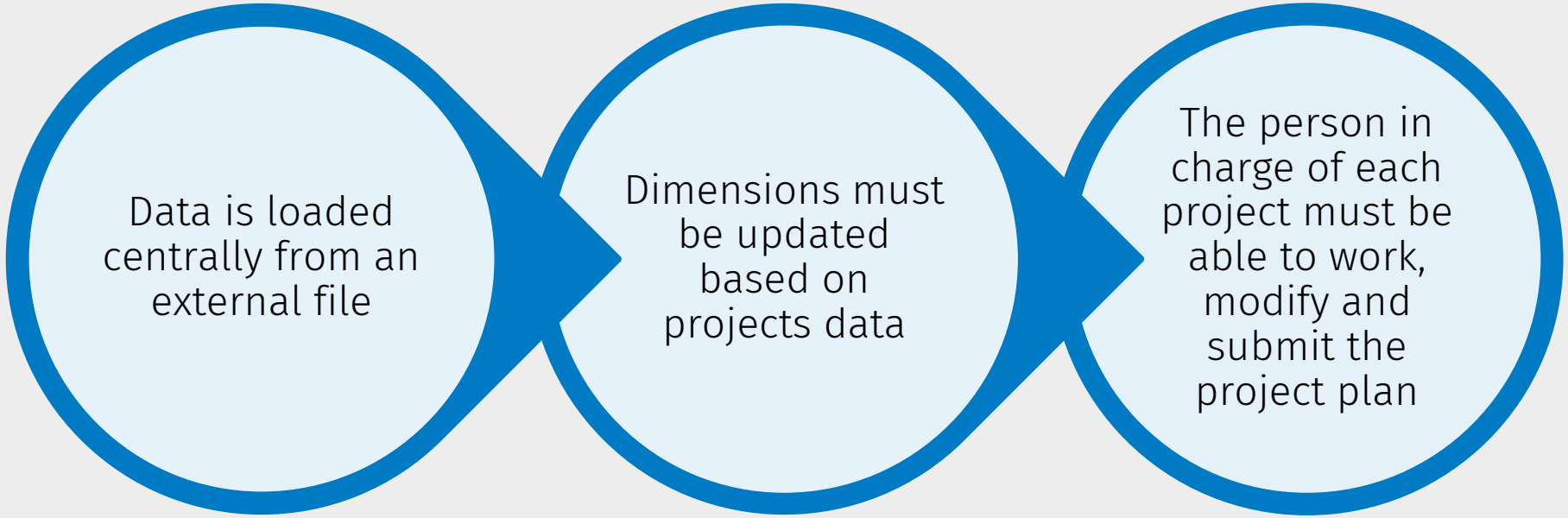
CONS

- Complex to setup
- Data processing run requires the use of an API Client

Let’s talk about: Business case – Data model update and budget pre-population



Requirements



Now it’s your turn

Exercise # 2



Data model update and budget pre-population - Solution



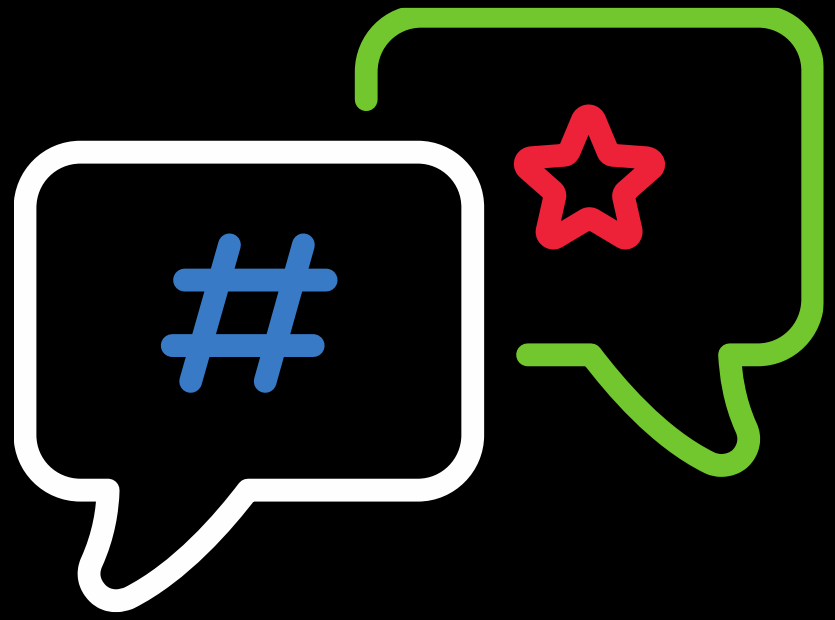
- A single DTP activity loads data and updates metadata
- An updated data model allows you to load new dimensional elements and their relationship in the hierarchy

Now it’s your turn

Exercise # 3



Let’s talk about: DTP performances



Drivers of performance

Performance is driven by

- Type and complexity of operations
- Number of table rows in calculations
- The order of operations may have a different impact on data processing time

What is the most efficient approach for perfomance?

Example: tables involved with different size

Price – (1.000 rows)

PRODUCT	PRICE	PRICE_CONVERTED
Antivirus	50	
Op.System	20	
Mouse	120	
…	…	

Quantity – (1.000.000 rows)

PRODUCT	CUSTOMER	QUANTITY
Antivirus	C1	10
Op.System	C1	20
Mouse	C1	30
…	…	…
Antivirus	C2	50
Op.System	C2	35
…	…	…



Target: Revenue

PRODUCT	**CUSTOMER**	**PRICE**	PRICE CONVERTED	QUANTITY	REVENUE

Options for calculation

Insert biggest dataset first

- Insert Quantity (1000000 rows)
- Enrich Price (1000 rows)
- Enrich with formula (Revenue)

Insert smallest dataset first

- Insert Price (1000 rows)
- Join Quantity (1000000 rows)
- Enrich with formula (Revenue)

Join and calculation performed in Virtual Dataset

- Virtual dataset (Price joins Quantity)
- DTP (Insert from another dataset)

Analysis of solutions



Insert biggest dataset first

- Slowest solution
- DTP takes a significant time to enrich 1.000.000 of rows

Insert smallest dataset first

- Faster than insert biggest dataset first
- Performance depends on the number of rows processed

Use of virtual dataset

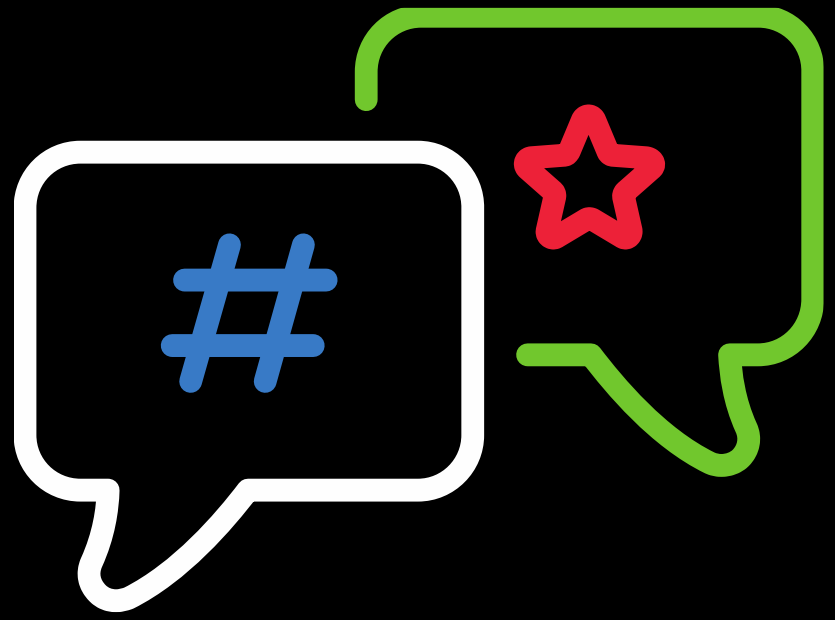
- Fastest solution
- All operations in one shot

Conclusions & suggestions



- Operations of Insert and Calculations are faster on a smaller number of rows
- Insert dataset and perform operations with less rows first
- Virtual dataset is faster but it has limits of use

Let’s talk about: Time frame management



Agenda

- Process types and process management
- Standard scenarios-periods vs year-month-week-day-version
- Multi-year horizon



Time related aspects

Multi-year horizon

- How to manage a time frame?
  - Data cross scenario period
  - Mapping
  - Coding

Structures beyond scenario-period

- Year-months
- Weeks-days

Particular requirements

- Versions

Managing time

- Period information in dedicated column
- Many amount columns representing different periods

Process number of cycles: scenario/period

Data is collected one month at a time, every period is a different cycle

- Same rules, same process
- AIH data lies in different partitions

Typical for month end closing and consolidation

ACTUAL process

- The process has no time reference
- All actual scenarios are linked to it
- Submission, locking, archiving are managed by single period

Process number of cycles: single

Collection of data

- At the same time for all the periods making part of the cycle.
- When the cycle changes, process must change.

Process number of cycles: single

Typical for planning and forecasting

- The process has time reference
- Submission and archiving are managed at once for all scenario-periods belonging to process
- Locking is managed by single period

Process number of cycles: single

Managing all scenarios in one process without time reference (e.g. BUDGET) is **not** a good practice:

- AIH data lays in the same partition: no partitioning benefits
- Archiving of old scenario-period is not possible
- Submission status of old data is lost
- Creation of scenarios-periods for the new cycle must be managed manually

Process number of cycles: single. Correct managament of scenarios / periods

Good practices:

Clone process functionality

- Use for a new cycle if the process configuration is the same
- You can also clone the related scenario(s)

Show end users only active processes

- Lock periods and archive data
- Delete processes and data no more required

In case of multi-year scenarios

- The version of process and scenario must have the information of the planning moment
- The coding can be used for adv. dynamics and calculations

Scenario-year

Scenario data could be classified in 2 analytical dimensions:

- Year (derivable also as scenario attribute)
- Version (text, scenario attribute too)

Year can be used:

- For calculations
- To avoid editability checks
- For some specific reports

Scenario must always be present:

- For multidimensional reports
- To leverage standard functionalities (time rollup, conversions, navigations, snapshots…)

Period

Period is meant to be the **month**

It’s not recommended to manage any other unit of time on period dimension.

Period coding allows **calculation**

Weeks or **days** can be managed in an analytical dimensions and aggregated through hierarchies

Management of time in different columns

Time on columns can be convenient for

- Data entry
- Calculations

For reporting purposes

It is advisable to unpivot and map information in scenario-period and the other possible time related dimensions.

Multi-years planning

How to manage time frame:

Opening many periods can cause inconveniences.

Not easy to:
- compare data
- understand dates

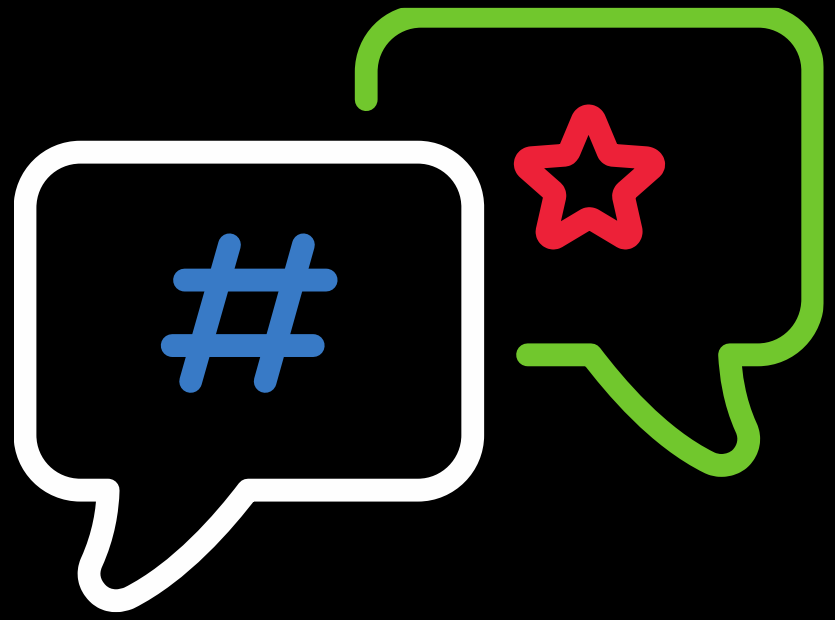
Managing scenario/period as they are meant to be is always a good idea

Multi-years planning scenario coding: example

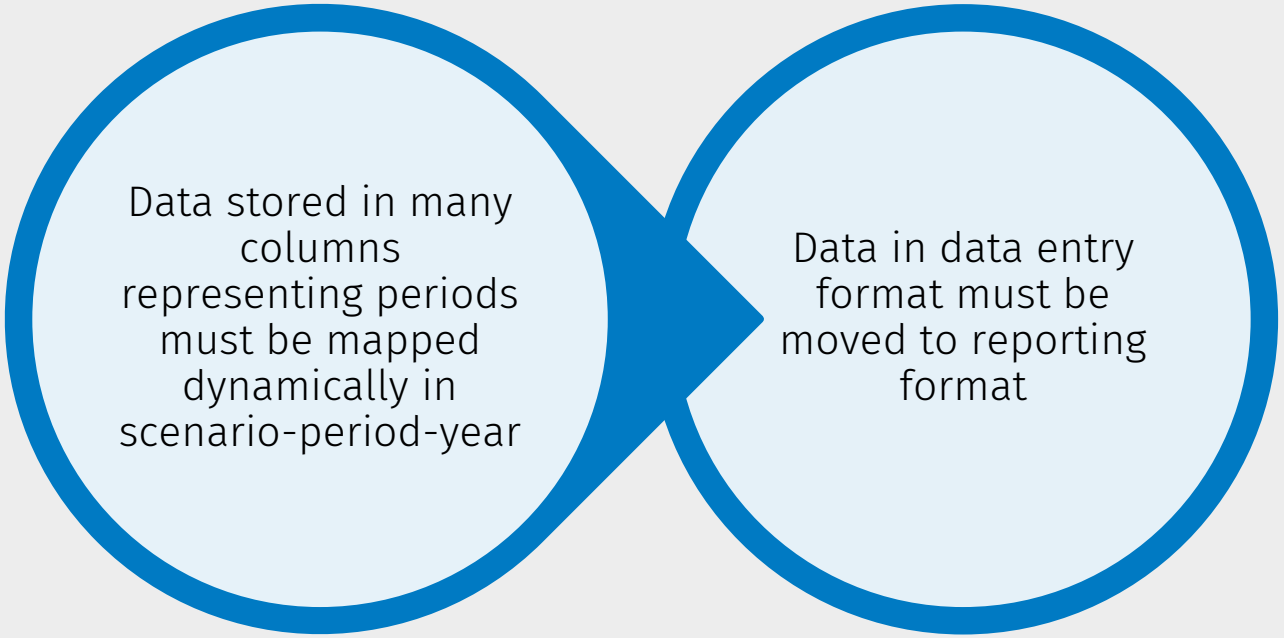
3-year planning done once a year	
2070_PLN70	2070 data planned in 2070
2071_PLN70	2071 data planned in 2070
2072_PLN70	2072 data planned in 2070
2071_PLN71	2071 data planned in 2071
2072_PLN71	2072 data planned in 2071
2073_PLN71	2073 data planned in 2071

2-year forecast done quarterly	
2070_F0370	2070 data forecasted in March 2070
2071_F0370	2071 data forecasted in March 2070
2070_F0670	2070 data forecasted in June 2070
2071_F0670	2071 data forecasted in June 2070
2070_F0970	2070 data forecasted in September 2070
2071_F0970	2071 data forecasted in September 2070
2071_F0371	2071 data forecasted in March 2071
2072_F0371	2072 data forecasted in March 2071

Let’s talk about: Business case – Multi-year planning and unpivoting periods



Requirements



Now it’s your turn

Exercise # 4



Multi-year planning and unpivoting periods - Solution



- Data entry structure with periods on columns is valid also cross year
- How to unpivot periods if time on columns is cross year?
- The traditional unpivoting opening period 1 to 12 and applying a formula has disadvantages
  - It does not include the information for different years
  - It requires a long formula

Solution

- A mapping for scenario/period based on the process
- Calculation based on enrichment with formula field

Unpivoting periods - example

Planning

Process	Dimensions	...	P11	P12	P13	P14	...
PLN70	XYZ		100	300	200	150	

Time mapping – reference dataset deploy

Process	Scenario	Period	Period field
PLN70	2070_PLN70	11	P11
PLN70	2070_PLN70	12	P12
PLN70	2071_PLN70	01	P13
PLN70	2071_PLN70	02	P14

Target – Staging area

Process	Dim.	P11	P12	P13	P14	Scenario	Period	Per. field	Amount

Unpivoting periods - example

Planning

Process	Dimensions	...	P11	P12	P13	P14	...
PLN70	XYZ		100	300	200	150	

Time mapping – reference dataset deploy

Process	Scenario	Period	Period field
PLN70	2070_PLN70	11	P11
PLN70	2070_PLN70	12	P12
PLN70	2071_PLN70	01	P13
PLN70	2071_PLN70	02	P14

- Insert planning
- Enrich with reference dataset (deploy)

Target – Staging area

Process	Dim.	P11	P12	P13	P14	Scenario	Period	Per. field	Amount
PLN70	XYZ	100	300	200	150	2070_PLN70	11	P11	
PLN70	XYZ	100	300	200	150	2070_PLN70	12	P12	
PLN70	XYZ	100	300	200	150	2071_PLN70	01	P13	
PLN70	XYZ	100	300	200	150	2071_PLN70	02	P14	

Unpivoting periods - example

Planning

Process	Dimensions	...	P11	P12	P13	P14	...
PLN70	XYZ		100	300	200	150	

Time mapping – reference dataset deploy

Process	Scenario	Period	Period field
PLN70	2070_PLN70	11	P11
PLN70	2070_PLN70	12	P12
PLN70	2071_PLN70	01	P13
PLN70	2071_PLN70	02	P14

Enrich with formula field
- Formula: Period field
- Target: Amount

Target – Staging area

Process	Dim.	P11	P12	P13	P14	Scenario	Period	Per. field	Amount
PLN70	XYZ	100	300	200	150	2070_PLN70	11	P11	100
PLN70	XYZ	100	300	200	150	2070_PLN70	12	P12	300
PLN70	XYZ	100	300	200	150	2071_PLN70	01	P13	200
PLN70	XYZ	100	300	200	150	2071_PLN70	02	P14	150

Unpivoting periods - example

Target – Staging area

Process	Dim.	P11	P12	P13	P14	Scenario	Period	Per. field	Amount
PLN70	XYZ	100	300	200	150	2070_PLN70	11	P11	100
PLN70	XYZ	100	300	200	150	2070_PLN70	12	P12	300
PLN70	XYZ	100	300	200	150	2071_PLN70	01	P13	200
PLN70	XYZ	100	300	200	150	2071_PLN70	02	P14	150

Target- Reporting dataset

Process	Dimensions	Scenario	Period	Amount
PLN70	XYZ	2070_PLN70	11	100
PLN70	XYZ	2070_PLN70	12	300
PLN70	XYZ	2071_PLN70	01	200
PLN70	XYZ	2071_PLN70	02	150

Copy only the relevant fields

Unpivoting periods - conclusions

Calculation is done in the project plan dataset

Its structure is more suitable and the reporting dataset structure is kept clean

Time mapping

Is a reference dataset deployed to open rows, process is key (condition)

Enrichment with formula fields

Allows you to avoid more complex formulas

Flexibility

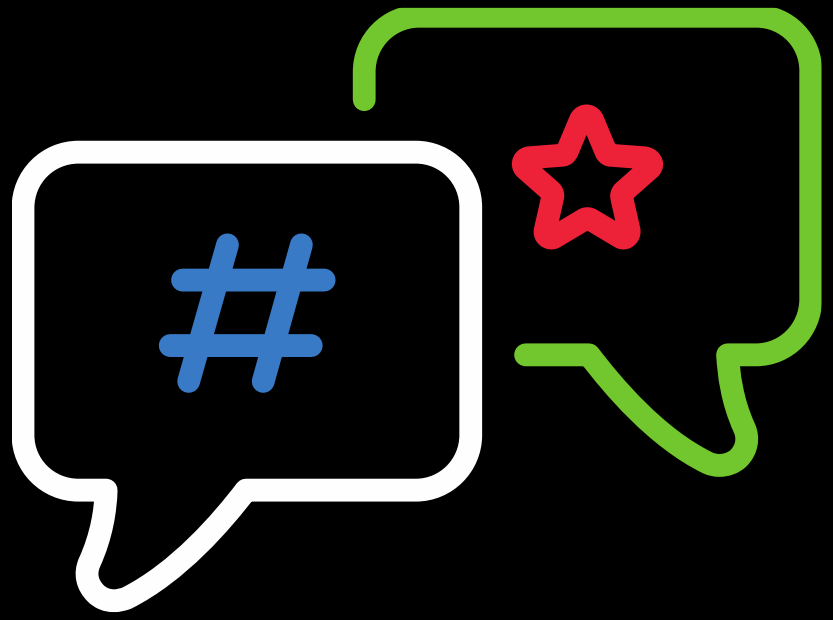
This solution gives the flexibility to move from an open structure with N periods to the precise definition of time, based on process

Now it’s your turn

Exercise # 5



Let’s talk about: Comparison with other data



Why is it important to compare data?

Is the essential property for an evaluation

Enables to judge the relevance of a result

Why is it important to compare data?

Comparing budget and actual figures (variance analysis) allows you to:

- Analyse real performance related expectations
- Understand the reason for the variance
- Identify ways to increase profitability and mitigate risks
- Take more informed decisions
- Check if the plan is still valid or if corrections and adjustments are necessary

A common reporting format

A common reporting structure is essential to facilitate comparisons

Comparisons between actual, budget, target and forecast can be made on the intersection of the dimensions of the scenarios considered

Allocations of data can increase the possible comparable data

Reporting tables update

Reporting tables may be updated:

- Straightaway after any data entry
- At the end of all the data collection process
- Any combination of the above

Reporting in Financial workspace

PROS	CONS
• Standardized structure • Out of the box tables • FW mapping to move data	• No analytical dimension details • Table size grows with data addition as time passes

Reporting in Analytical workspace

PROS	CONS
• All dimensions available • Partitioning benefits	• Custom data processing to move data

Reporting – Query

The creation of reports using query to retrieve data is not a good practice:

- No leverage of built-in functionalities
- Hard to hand over and to maintain

Use the matrix wizard!

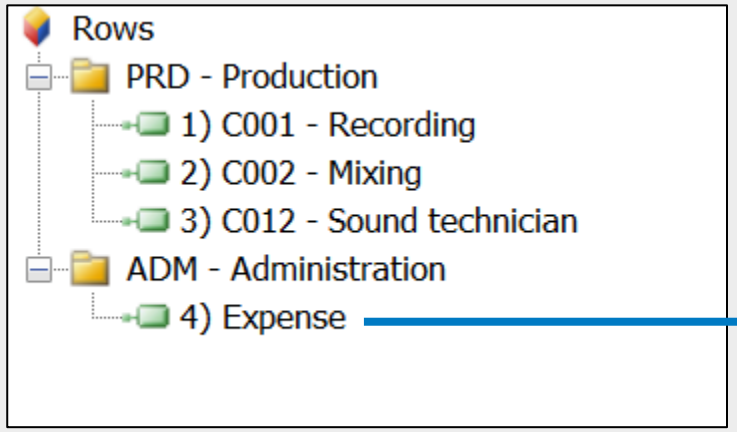
Reporting – Missing elements in dataset dimensional fields

SCENARIO	PERIOD	EXPENSE	AMOUNT
2070_PLN70	12	GEN	87.000000000
2070_PLN70	12		69.000000000
2070_PLN70	12	C001	23.000000000

Pay attention to missing elements in reporting datasets

Data does not appear in the roots of the:

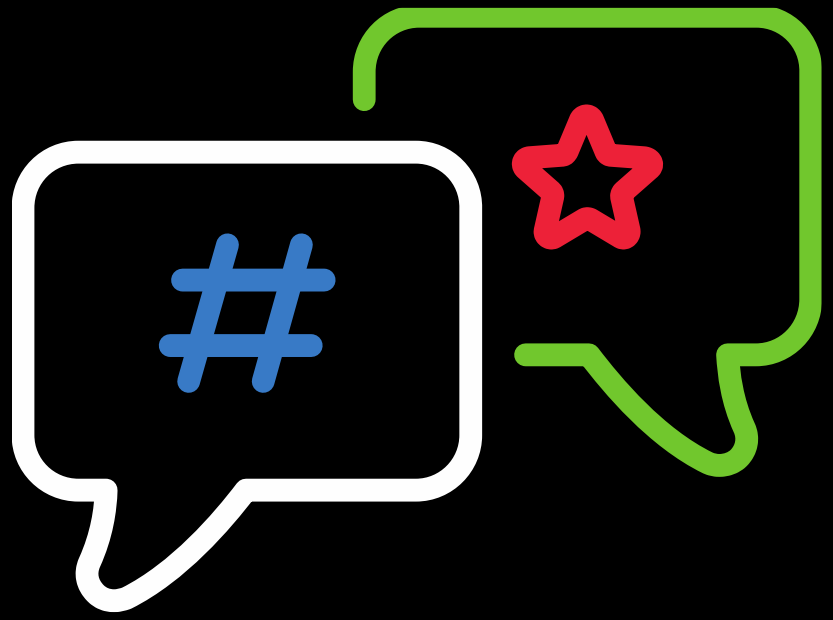
- Flat list
- Hierarchies



You will not see the amount of 69 with this filter

Element specification is a good practice!

Let’s talk about: Global assumptions



Assumptions application

Assumpions are:

- Factors of a calculation, specified at higher levels
- Applied to a more detailed list of elements, e.g :
  - Globally
  - By country
  - By product family
  - By customer class
  - By year

The challenge is:

- How to match the dimensionality of the assumption with all the elements for which it has been specified.

Assumptions in Analytic Information Hub

In relational database the problem is solved identifying the key dimensions that allow us to:

- Join the assumption
- Join the target data.

Loop on time calculation (data of a period depending on cascade from previous period) is complex.

Calculation in Excel can be used to prepare data for all periods.

Assumptions in Analytic Information Hub

Assumptions

- Can be managed in reference datasets (field type condition, attribute, level and filter)

The order of the operation may impact on performance:

- Insert dataset + enrichment with reference
- Insert with reference dataset
- Insert + join dataset
- Insert virtual dataset

Datasource or Insert data from other scenario-period

- can be used to retrieve assumption from other sources.



First do the operation involving less data!

Insert data from other scenario-period

For DTP operation of:

- Insert
- Enrichment
- Join
- Reading from partitioned dataset

It is possible to read data from other scenario-periods:

- Using the tab Advanced
- Scenario period can be referenced also dynamically from context



Less use of datasources

Datasources must still be used in particular cases (e.g. multiple scenarios related to process)

Assumptions in Financial workspace

Assumptions

- Are tipycally managed on generic elements or ad-hoc statistical accounts.

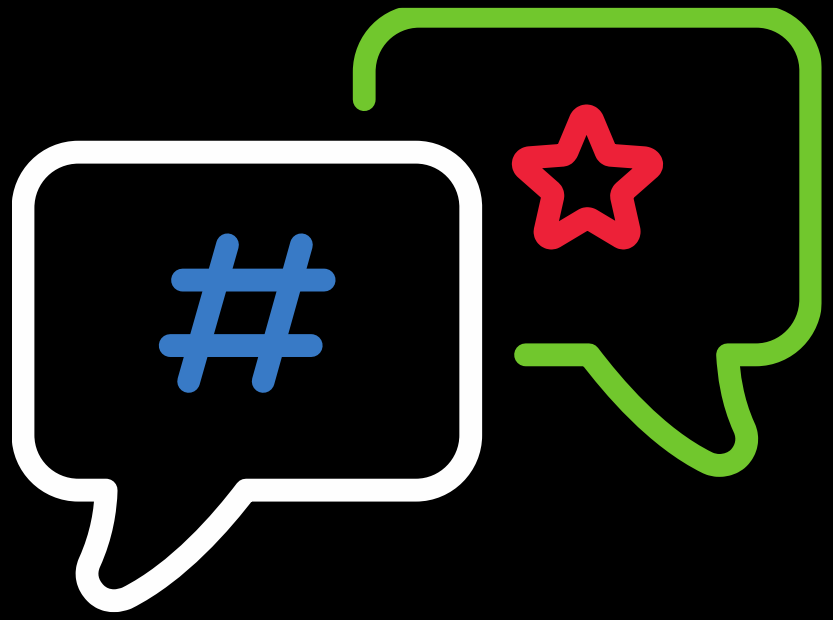
The application

- It is usually done with MDX calculation logics

Loop

- On time calculation is possible with a specific syntax

Let’s talk about: Forecast



Forecasting in CCH Tagetik

Analytic Information Hub

- Allows you to manage different data models suitable for any function involved in the forecasting process.

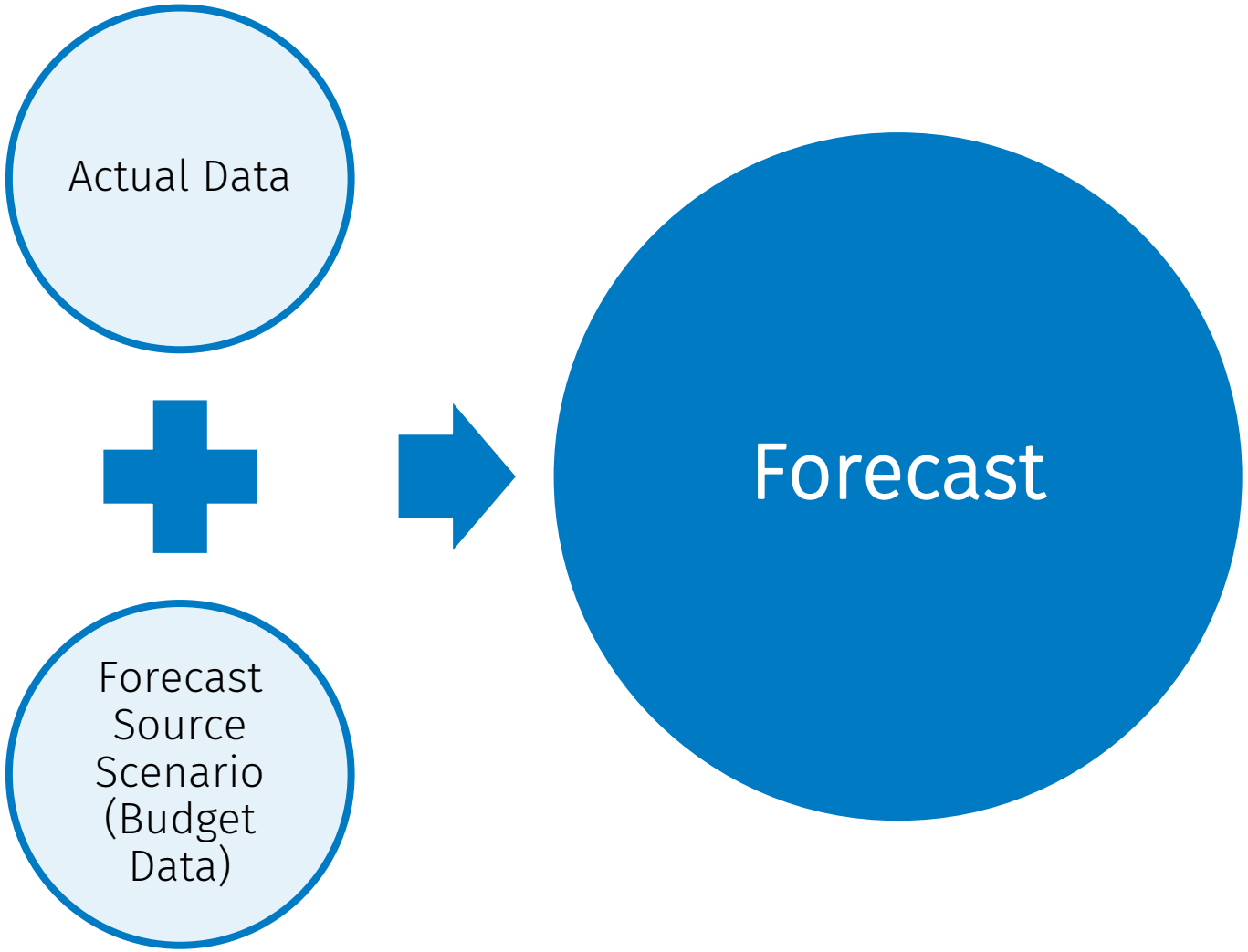
Predictive intelligence

- Provides estimates based on machine learning.
- Actual data allows us to establish relationships among variables that can be helpful to predict the future behavior of the values.

Financial workspace

- Provides built-in forecast models for financial forecasting.
- Essentially, it is a pre-population tool.

Forecast Models



Forecast models logics

Hold the to-year

- Keeps the monthly budget till year end
- Common when year end is close

Hold the annual total

- Keeps the annual amount budgeted, adjusting monthly data
- Common when year end is far

Rolling forecast and multi-year horizon

In case the forecast source data

Has same the time frame

- We can use the logics of the models within the year.
- Hold annual total becomes hold multi-year total.

Is available only for the first year

- The challenge is how to prefill the following years.

Forecast - Conclusions

The main challenges are

- The possible mismatches between actual and planning data
- The logics of pre-population and data processing

Points of attention

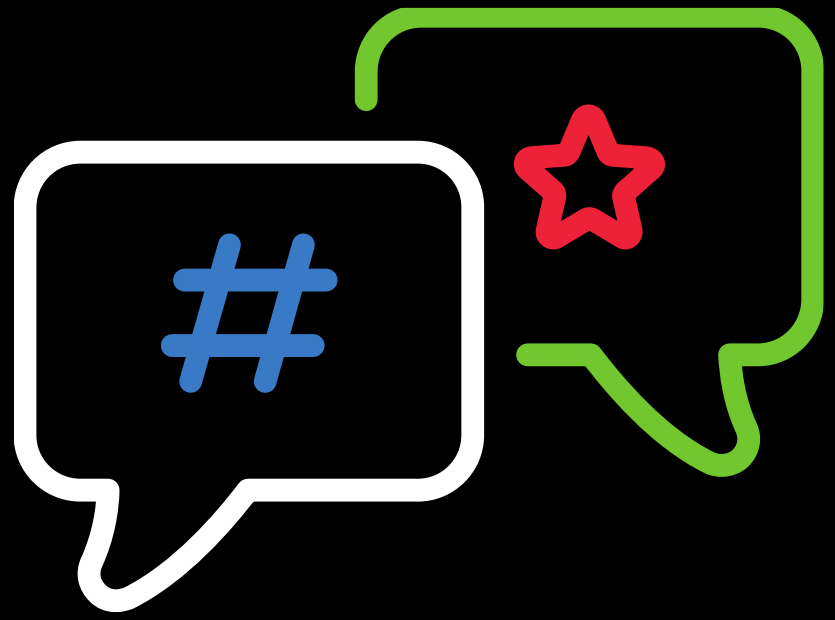
- Time management for rolling forecast can be managed by process with the same logics of multi-year planning
- Monthly forecast, even within the year, requires a new process and a new scenario each month (if previous forecast data is needed)

In AIH

- We can recreate the forecast models present in Financial workspace
- Data saved monthly makes easier the calculation

Let’s talk about:

Business case – Planning adjustment and forecast



Requirements

Planning

- Is compared with last available actual at actual dimensionality

Forecast baseline

- Is produced spreading proportionally in the forecast periods the remaining budget not yet spent

Baseline

- Is revaluated with compound inflation rates

Forecast adjusted

- Is distributed to all available planning dimensions

Planning adjustment and forecast - analysis



The problem can be split in the following questions

- How to allocate actual data on singers and expenses?
- How to calculate the coefficient to spread proportionally on forecast periods the current variance between actual and planning?
- How to calculate the compound inflation rate?

It is not feasible to read data from other scenario/period using the advanced setting of DTP.

In a multi-year process the partition is not identified by a single scenario.

Now it’s your turn

Exercise # 6



Planning adjustment and forecast - solutions



How to allocate actual data on singers and expenses?

- DTP allocations. Data to allocate and drivers read with datasource.

How to calculate the coefficient to spread proportionally on forecast periods the current variace between actual and planning?

- Process grouped data through DTP in dedicated dataset fields. As the processing involves less data, it will be done first.

How to calculate the compound inflation rate?

- Calculation is done in excel and stored in a reference dataset.

All calculations are performed in a specific dataset. Result is copied in reporting dataset.

Planning adjustment and forecast – Coefficient to hold the total

Reporting dataset – Actual data

Scenario	Period	Project	Country	Account	Amount
2070ACT	07	P001	ITA	50888	100
2070ACT	08	P001	ITA	50888	50
2070ACT	09	P001	ITA	50888	200
2070ACT	10	P001	ITA	50888	100

- Insert actual to date total by Project, Country, Account
- Sum amount in Act to date field

Target – Staging area

Scenario	Period	Project	Country	Account	Amount	Tot project	Act to date	Pln to date	Coefficient
		P001	ITA	50888			450		

Planning adjustment and forecast – Coefficient to hold the total

Reporting dataset – Planning data

Scenario	Period	Project	Country	Account	Singer	Expense	Amount
2070PLN70	07	P001	ITA	50888	S001	C001	80
2070PLN70	08	P001	ITA	50888	S001	C002	50
2070PLN70	09	P001	ITA	50888	S002	C001	170
2070PLN70	10	P001	ITA	50888	S002	C003	90

- Join planning to date, total by Project, Country, Account
- Sum amount in Pln to date field

Target – Staging area

Scenario	Period	Project	Country	Account	Amount	Tot project	Act to date	Pln to date	Coefficient
		P001	ITA	50888			450	390	

Planning adjustment and forecast – Coefficient to hold the total

Reporting dataset – Planning data

Scenario	Period	Project	Country	Account	Singer	Expense	Amount
2070PLN70	07	P001	ITA	50888	S001	C001	80
2070PLN70	12	P001	ITA	50888	S001	C002	120
2071PLN70	04	P001	ITA	50888	S002	C001	350
2072PLN70	02	P001	ITA	50888	S002	C003	60

- Join **planning total** by Project, Country, Account
- Sum amount in Total project field

Target – Staging area

Scenario	Period	Project	Country	Account	Amount	Tot project	Act to date	Pln to date	Coefficient
		P001	ITA	50888		**1140**	450	390	

Planning adjustment and forecast – Coefficient to hold the total

- Enrichment with formula: **coefficient** calculation
- (Tot project – Act to date) / (Tot project – Pln to date)

Target – Staging area

Scenario	Period	Project	Country	Account	Amount	Tot project	Act to date	Pln to date	Coefficient
		P001	ITA	50888		1140	450	390	0.92

Planning adjustment and forecast – Forecast periods pre-population

Reporting dataset – Planning data

Scenario	Period	Project	Country	Account	Singer	Expense	Amount
2070PLN70	07	P001	ITA	50888	S001	C001	80
2070PLN70	12	P001	ITA	50888	S001	C002	120
2071PLN70	04	P001	ITA	50888	S002	C001	350
2072PLN70	02	P001	ITA	50888	S002	C003	60

- Join **planning periods** by Project, Country, Account
- Filter on future periods
- Attribute: Scenario, Period, Singer, Expense, Amount
- Scenario is updated

Target – Staging area

Scenario	Period	Project	Country	Account	Singer	Expense	Amount	Coefficient
2070F1070	12	P001	ITA	50888	S001	C002	120	0.92
2071F1070	04	P001	ITA	50888	S002	C001	350	0.92
2072F1070	02	P001	ITA	50888	S002	C003	60	0.92

Planning adjustment and forecast – Inflation

Reference dataset – Inflation

Scenario	Period	Country	Infl rate	Comp rate
2070F1070	12	ITA	0.1	1.1
2071F1070	04	ITA	0.1	1.21
2072F1070	02	ITA	0.1	1.331

- Enrichment with reference dataset
- Attribute: compound inflation rate

Target – Staging area

Scenario	Period	Project	Country	Account	Singer	Expense	Amount	Coefficient	Comp rate
2070F1070	12	P001	ITA	50888	S001	C002	120	0.92	**1.1**
2071F1070	04	P001	ITA	50888	S002	C001	350	0.92	**1.21**
2072F1070	02	P001	ITA	50888	S002	C003	60	0.92	**1.331**

Planning adjustment and forecast – Forecast periods calculation

- Enrichment with formula: apply coefficient and inflation
- Amount = Amount x coefficient x comp rate

Target – Staging area

Scenario	Period	Project	Country	Account	Singer	Expense	Amount	Coefficient	Comp rate
2070F1070	12	P001	ITA	50888	S001	C002	121.44	0.92	1.1
2071F1070	04	P001	ITA	50888	S002	C001	389.62	0.92	1.21
2072F1070	02	P001	ITA	50888	S002	C003	73.47	0.92	1.331

Planning adjustment and forecast – Actual periods

Actual data is input in forecast scenario

- Reading from reporting dataset
- Using planning data for the same periods as driver to allocate actuals on:
  - Singers
  - Expenses



The forecast is ready!

Planning adjustment and forecast – Copy on reporting

To copy data on reporting dataset there are two options:

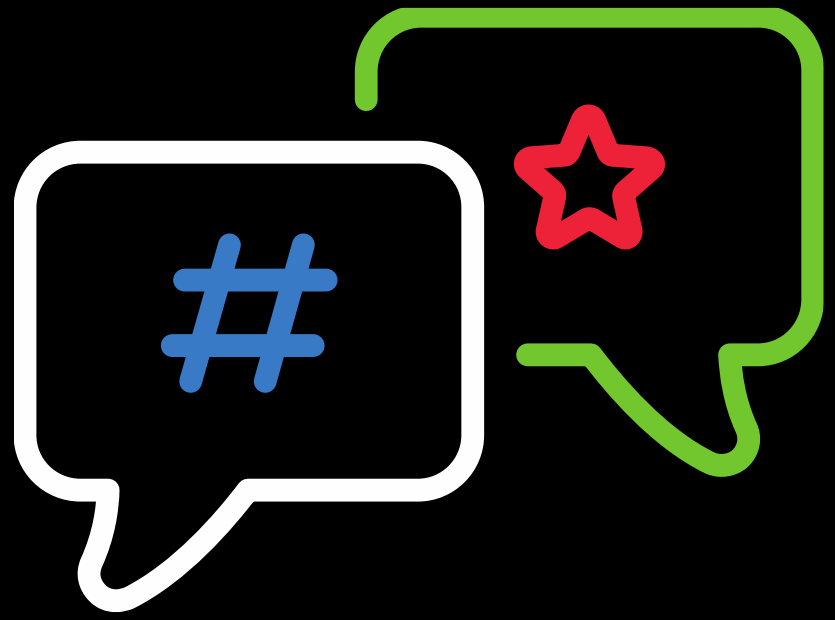
A new DTP activity	Loading defined on last DTP activity:
• Target dataset: 20 Reporting • Update reporting layer • Operation: insert from another dataset • Source dataset: 30 Forecast • Typology: Calculation	• Write to another dataset • Datasource mapping defined on dataset 20 Reporting
Faster definition if dataset fields are coherent No checks during update	Definition outside DTP QDL editability checks

Now it’s your turn

Exercise # 7



Let’s talk about: Financial Workspace tools

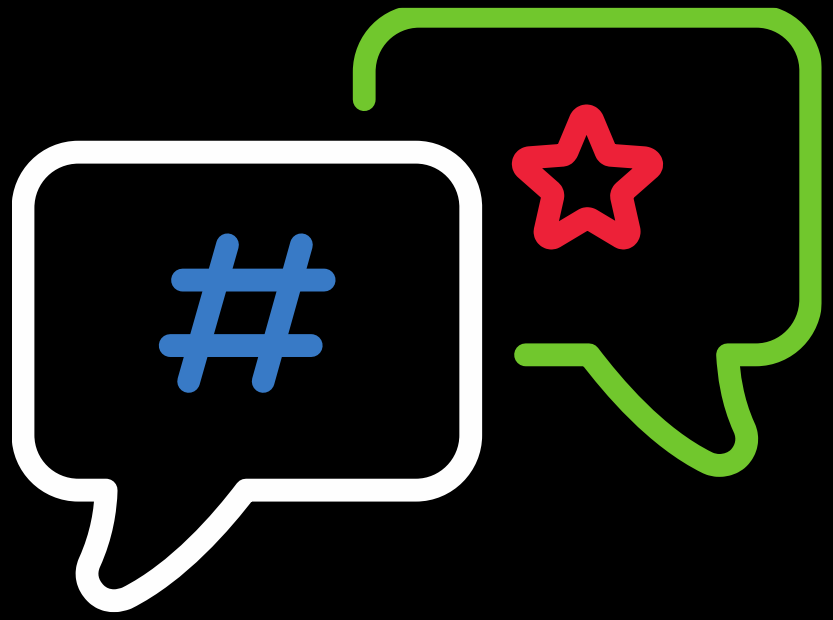


Agenda

- MultiDimensional calculation logic in a nutshell
- Advanced FST & KPI



Let’s talk about: Multidimensional Calculation Logics in a nutshell



Multi Dimensional Calculation Logics features

MDX-like scripting language

- TGKML
- High performance tool for complex calculations

Cubes

- Represent input and output tables in the script
- Read and write on any original and cosolidation FW table

Dimensions

- All financial dimensions
- Possibility to filter and read from fixed point
- Applicable to both input and output data

Time movements

- To read data on Scenario/Period different from running filters
- Movements: Forward-Backward, Upward-Downward, Parallel period.

P X Q Example

```
[LO].[REVENUE]:= [PRICE] * [QUANTITY];
```

Account	Cost Center	Product	Amount
PRICE	SALES IT	Antivirus	10
PRICE	SALES USA	Mouse	20

Account	Cost Center	Product	Amount
QUANTITY	SALES IT	Antivirus	5
QUANTITY	SALES USA	Mouse	10
~~QUANTITY~~	~~SALES UK~~	~~Scanner~~	~~8~~

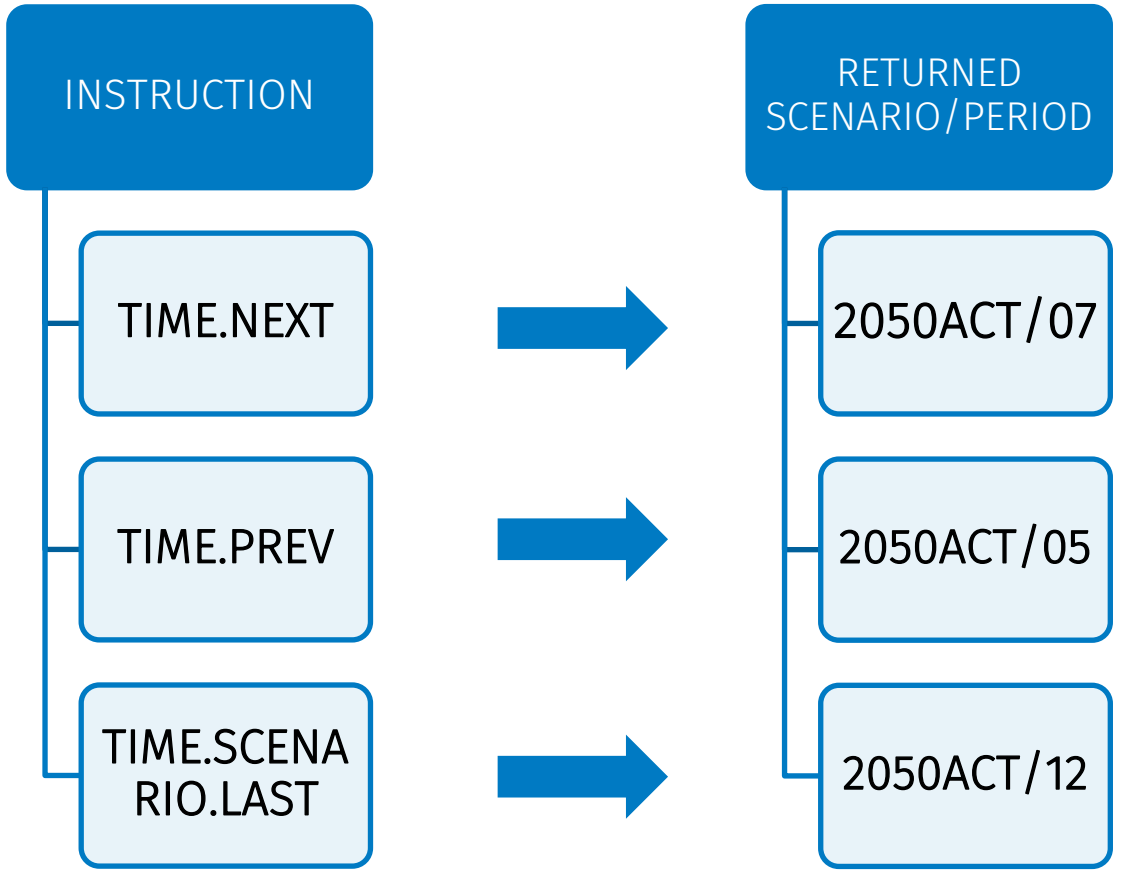


2 Rows in Output

Account	Cost Center	Product	Amount
REVENUE	SALES IT	Antivirus	10 * 5 = 50
REVENUE	SALES USA	Mouse	20 * 10 =200

Time Movements - Examples

Assume the script is run for 2050ACT/06



P X Q example with output Account filter

```
[LO].[REVENUE] (Dest2.Filter([Modem],[Scanner]):= [PRICE] * [QUANTITY];
```

Account	Cost Center	Product	Amount
PRICE	SALES IT	Modem	10
PRICE	SALES USA	Scanner	20
PRICE	SALES_UK	Mouse	30

Account	Cost Center	Product	Amount
QUANTITY	SALES IT	Modem	5
QUANTITY	SALES USA	Scanner	10
QUANTITY	SALES UK	Mouse	8



2 Rows in Output

Account	Cost Center	Product	Amount
REVENUE	SALES IT	Modem	10 * 5 = 50
REVENUE	SALES USA	Scanner	20 * 10 =200

LoopOnTime

When amount of a period depends on the same account in the previous period, we can use the `LoopOnTime` syntax

```
[LO].[50010](LoopOnTime):=[50010](Time.Prev)
```

Pros and Cons of Calculation logic

Pros

- High performance tool
- High level of customization

Cons

- Not user friendly
- High technical skills required

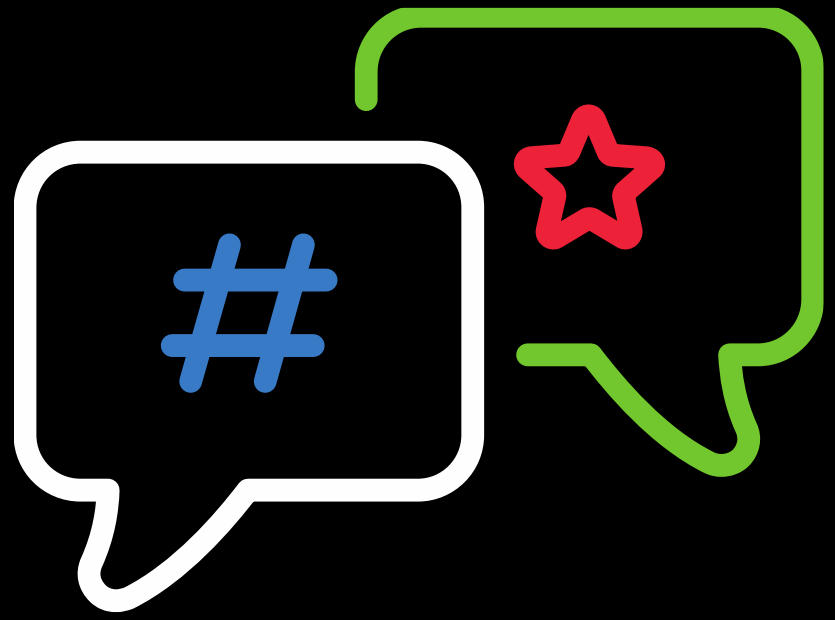
Planning adjustment and forecast – alternative solutions



- Planning and actual comparison made in FW
- Forecast model run with standard tool
- Inflation applied through mdx calculation
- Allocation to singer and expenses done moving data to AIH

- **Pro:** faster configuration using built-in tools
- **Cons:** Use of different tools, deeper knowledge

Let’s talk about: Advanced FST & KPI



FST Standard and advanced

FST

- Alternative aggregation structure available for account dimension
- Can have only 2 levels

FST Standard: features

Level 1: FST Item:

- Parent item: Node where 1 or more accounts relate
- Formula
- Description

Level 2: Account

FST Advanced: features

Level 1: FST item:

- Parent item: Node where a combination of Account/Custom dimension relate to provide additional detail on the dimension enabled
- Formula and description as in standard FST

Level 2: Account/Custom dimension

FST Advanced

When to use?

- When a schema depends on a Custom dimension

How to define?

- Enable the custom dimension in FST &KPI dimensionality
- Define the relationships between FST item and Account/Custom dimension

Which outcome to expect?

- FST items are calculated as the sum of the Account/Custom dimension elements combination

FST Advanced: Definition and Reporting outcome

Relationships Definition

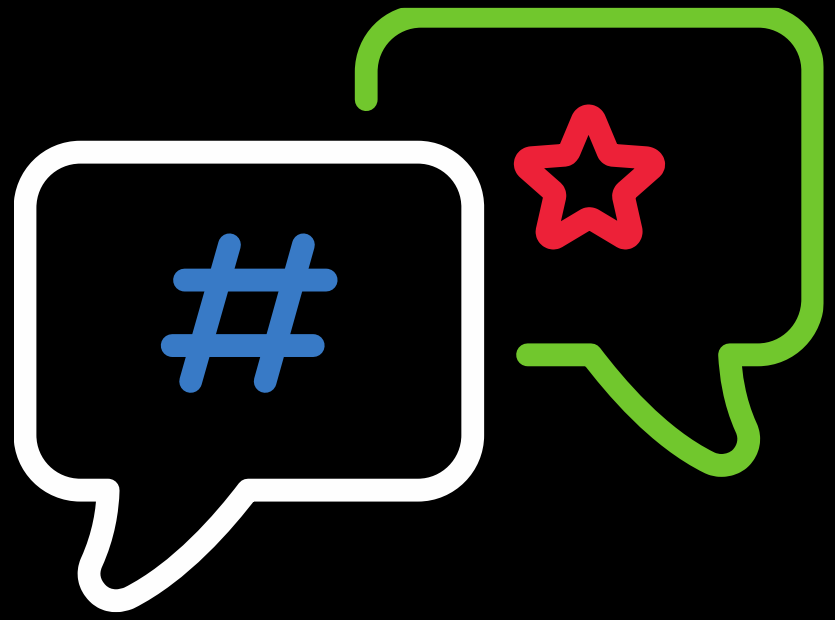
FST item	Account	Product (CD2)
010 - Revenue Software	50010 - Gross Revenues	01|X01 - Software
011 - Revenue Hardware	50010 - Gross Revenues	01|X02 - Hardware
012 - Revenue Electronics	50010 - Gross Revenues	01|X03 - Electronics
020 - Cost of good sold disc	50110 - Cost of Good Sold	A01 - Discs
021 - Cost of good sold books	50110 - Cost of Good Sold	A02 - Books
030 - Personnel expenses	02|02 - Personnel expenses	GEN - GEN

Advanced FST in matrix

	2052 Budget
Revenue Software	78.700,00
Revenue Hardware	56.700,00
Revenue Electronics	47.800,00
Cost of good sold disc	3.800,00
Cost of good sold books	4.500,00
Personnel expenses	44.900,00
Margin	130.000,00

FST item can be inserted only as Synthetic (not Analytic)

Let’s talk about: Allocations (optional)



Allocations

Why allocations are important?

To distribute a top-down adjustment value to other dimensions

To allocate a value stored by generic elements to specific elements

To charge generic costs and assess product profitability

To assign resources to support organization goals

Allocations in CCH Tagetik

Analytic Information Hub

- DTP standard operation for allocations
- Custom data processing

Financial allocations

Built-in module that allows you to:

- Create **double-entry allocation journals**, using both financial and non-financial drivers
- **Adjust in an automatic** way data coming from the general ledger or other systems

Allocations in Analytical Information Hub

PROS

- DTP dedicated operation
- Suitable for massive allocations
- FW mapping to move data

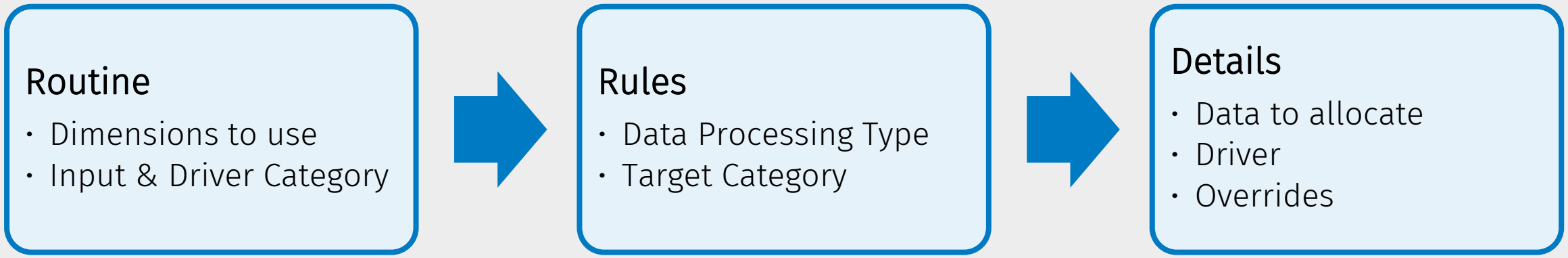
CONS

- Only to distribute amounts based on drivers
- Write-off not managed
- Limits for allocation cross scenario-period-entity

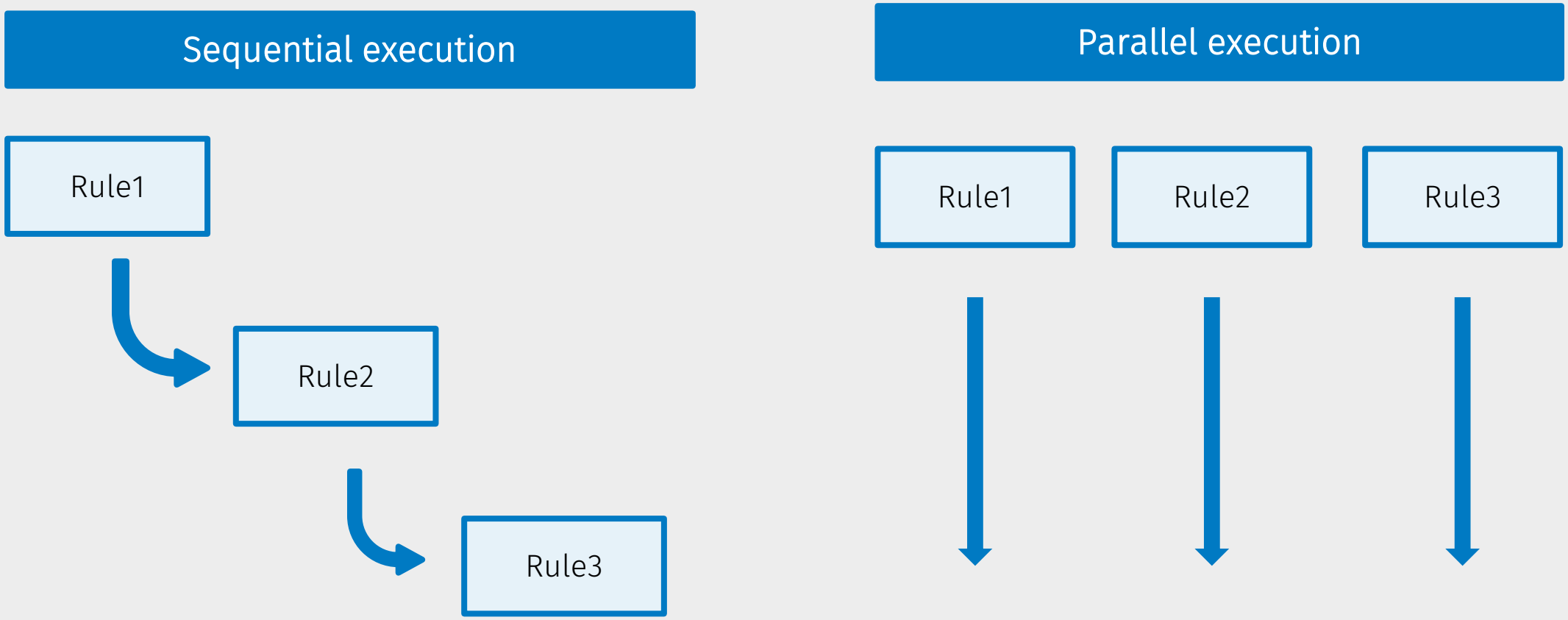
Financial Allocations

PROS	CONS
• A rich module with many options • No scripting • Synthetic definition through hierarchies • Write-off and data tracking	• Not suitable for massive allocations • No analytical dimensions

Allocations in CCH Tagetik Financial workspace

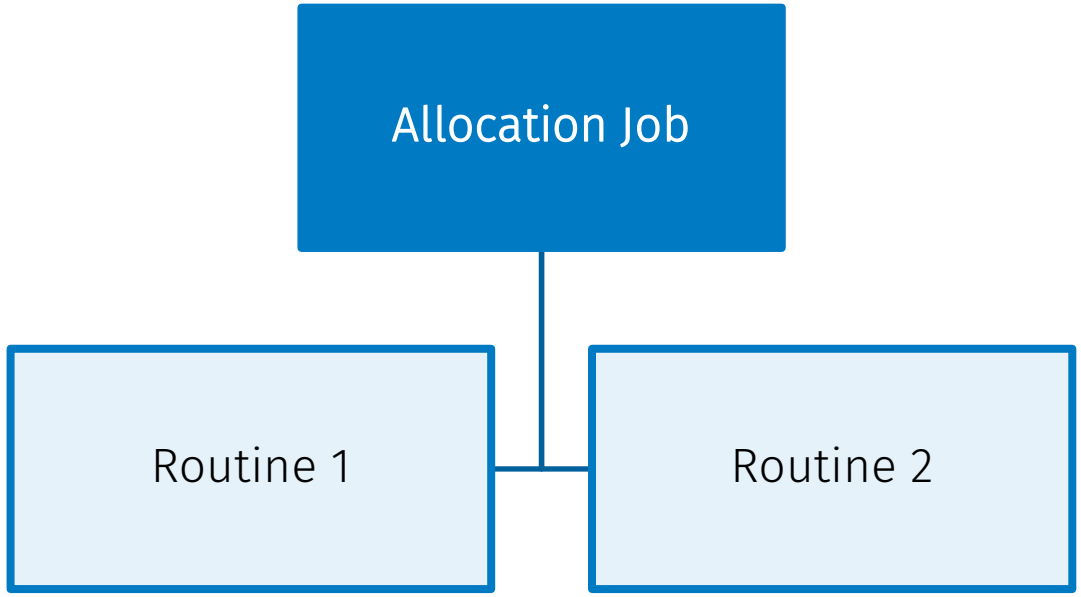


Structure - Rules



Allocation jobs

Allocation Jobs can contain many routines and can be scheduled



Data processing types

Allocation

Distribute amounts on other dimensional elements based on a driver

- **E.g.** from Costcenter01 to Costcenter02 and Costcenter03

Account allocation

Distribute amounts on the same dimensional combination to other account

- **E.g.** from Account01 to Account02 and Account03

Partial allocation

Same as allocation, but excluding elements

- **E.g.** excluding Costcenter GEN

Reclassification

Amounts are entirely moved to another dimensionality

- **E.g.** from Product01 to Product02

Allocation: rows generated

Data to allocate			
Account	Cost Center	Category	Amount
Energy	ADM	$AMOUNT	1000

Driver			
Account	Cost Center	Category	Amount
Headcount	ADM	$AMOUNT	10
Headcount	SLS	$AMOUNT	15
Headcount	MRK	$AMOUNT	25
Headcount	PRD	$AMOUNT	50

Rows generated by Allocation				
Account	Category	Cost Center	Amount	Entries
Energy	CAT_ALL	ADM	-1000	Write off
Energy	CAT_ALL	ADM	100	Allocated
Energy	CAT_ALL	SLS	150	Allocated
Energy	CAT_ALL	MRK	250	Allocated
Energy	CAT_ALL	PRD	500	Allocated

Simple Example

Rent Expenses Allocation to Cost Center

Sender

Cost Center	Account	Amount
Shared Service	**Rent**	**5,000**

Driver

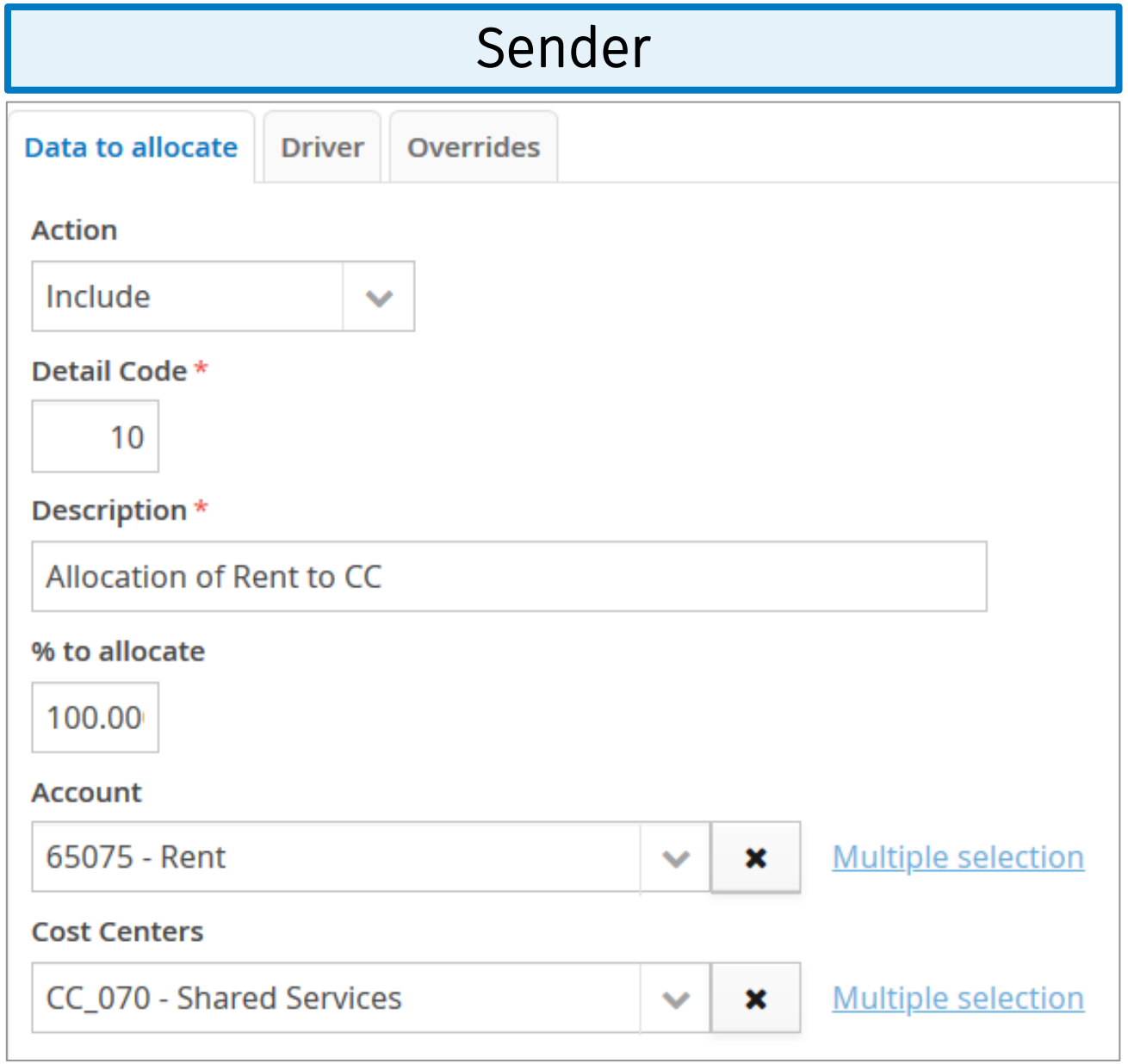
Cost Center	Account	Amount
Admin	**Square Meters**	10
Sales		30
Marketing		40
Total		**80**

Receivers

Cost Center	Account	Allocated Amount	Total Amount
Shared Service	**Rent**	-5,000	0
Admin		625	625
Sales		1,875	1,875
Marketing		2,500	2,500
Total			**5,000**

Write Off

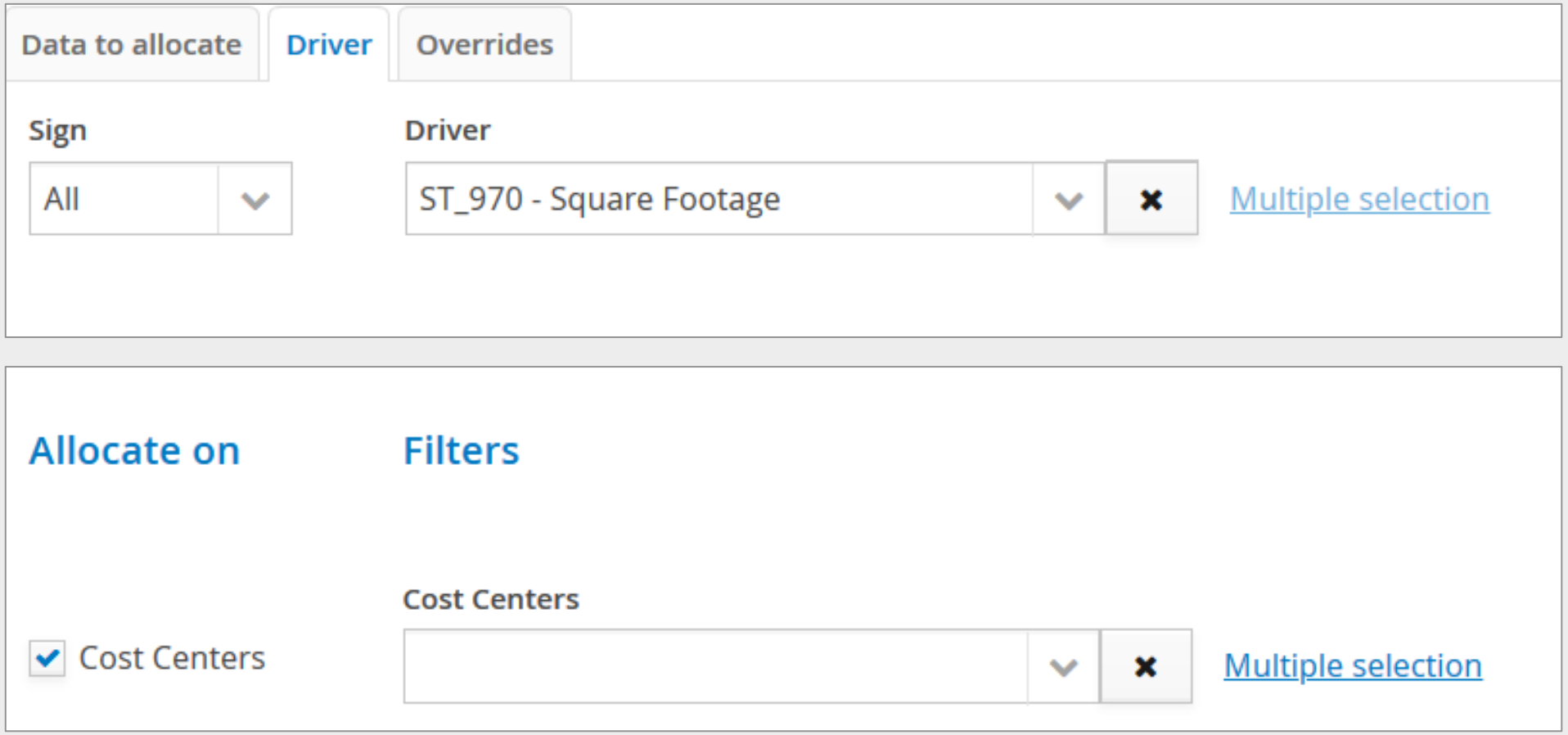
Simple Example – Configuration



Simple Example – Configuration

Driver

Receiver



A more complex case

Department Rent Expenses Allocation to Cost Center

Sender

Rent

Department	Cost Center	Amount
Sales & Marketing	S&M Shared Services	**5,000**
Admin & HR	Marketing	
	Sales	
	Admin & HR Shared Services	**7,000**
	Admin	
	HR	
Total		**12,000**

Allocation must occur for each department

A more complex case

Department Rent Expenses Allocation to Cost Center

Driver

Square Footage		
	Cost Center	Amount
Sales & Marketing	Marketing	20
	Sales	30
Subtotal		**50**
Admin & HR	Admin	60
	HR	80
Subtotal		**140**
Total		**190**

Driver Total will be considered for each department

A more complex case

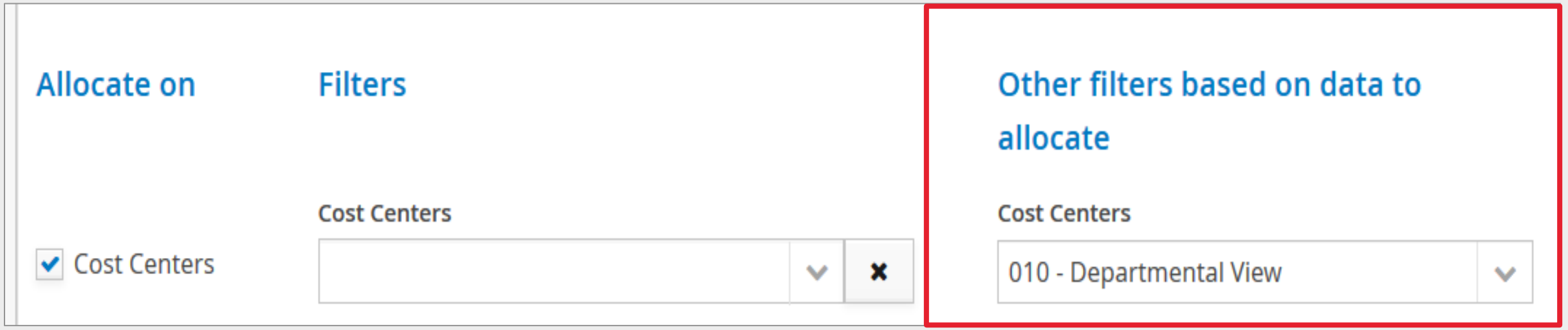
Department Rent Expenses Allocation to Cost Center

Receiver

Rent			
Department	Cost Center	Allocated Amount	Total Amount
Sales & Marketing	S&M Shared Services	-5,000	0
	Marketing	1,250	2,000
	Sales	1,875	3,000
Subtotal			**5,000**
Admin & HR	Admin & HR Shared Services	-7,000	-7,000
	Admin	3,000	3,000
	HR	4,000	4,000
Subtotal			**7,000**
Total			**12,000**

Allocation occurs based on department values

A more complex case - How can we achieve the result in the example?



Used to filter the elements on which the allocation will occur

Used to decide whether to group or not. Allocation must occur within these elements. It is possible to select the hierarchy to use for the grouping

Differences in result: No grouping

Department	Cost Center	Percentage Split	Allocated Amount	Total Amount
Sales & Marketing	S&M Shared Services			0
	Marketing	10.53%	1,263	1,263
	Sales	15.79%	1,895	1,895
Subtotal				**3,158**
Admin & HR	Admin & HR Shared Services			0
	Admin	31.58%	3,789	3,789
	HR	42.11%	5,053	5,053
Subtotal				**8,842**
Total		**100%**		**12,000**

Calculation on All dimension

Allocation on All Elements

Differences in result: Grouping on Department

Department	Cost Center	Percentage Split	Allocated Amount	Total Amount
Sales & Marketing	S&M Shared Services		-5,000	0
	Marketing	40%	2,000	2,000
	Sales	60%	3,000	3,000
Subtotal		**100%**		**5,000**
Admin & HR	Admin & HR Shared Services		-7,000	-7,000
	Admin	43%	3,000	3,000
	HR	57%	4,000	4,000
Subtotal		**100%**		**7,000**
Total				**12,000**

Calculation Weighed by node

Allocation split according to node

Some More Complex Examples - Data to be allocated on Multiple Dimensions

Data to Allocate:
Marketing Cost by product

LoB1	2,000	Pr1	1,000
		Pr2	1,000
LoB2	3,000	Pr3	2,000
		Pr4	1,000

Driver:
Revenue by Product & Customer

				Tier1		Tier2	
				1,400		1,300	
				Cu1	Cu2	Cu3	Cu4
				900	500	700	600
LoB1	1,200	Pr1	800	100	200	300	200
		Pr2	400	100	100	100	100
LoB2	1,500	Pr3	1,000	500	100	200	200
		Pr4	500	200	100	100	100

Some More Complex Examples

Case 1 – Allocate on Customer without grouping

	Alloc.	Filter
Customer	Y	
Product		

Calculate One set of Drivers per Customer

Driver	33%	19%	26%	22%

Don't Consider Product when calculating the Driver

				Tier1		Tier2		
				1,400		1,300		
				Cu1	Cu2	Cu3	Cu4	Total
				900	500	700	600	2,700
LoB1	1,200	Pr1	800	100	200	300	200	
		Pr2	400	100	100	100	100	
LoB2	1,500	Pr3	1,000	500	100	200	200	
		Pr4	500	200	100	100	100	

Some More Complex Examples

Case 1 – Allocate on Customer without grouping

	Alloc.	Filter
Customer	Y	
Product		

Driver	33%	19%	26%	22%

Allocation on Single Driver Calculated Previously

LoB1	2,000	Pr1	1,000
		Pr2	1,000
LoB2	3,000	Pr3	2,000
		Pr4	1,000

Tier1		Tier2	
2,593		2,407	
Cu1	Cu2	Cu3	Cu4
1,667	926	1,296	1,111
1000*33%	1000*19%	1000*26%	1000*22%
1000*33%	1000%19%	1000*26%	1000*22%
2000*33%	2000*19%	2000*26%	2000*22%
1000*33%	1000*19%	1000*26%	1000*22%

Some More Complex Examples

Case 2 – Allocate on Customer grouping on Product elements

	Alloc.	Filter
Customer	Y	
Product		Elem

				Cu1	Cu2	Cu3	Cu4
				900	500	700	600
LoB1	1,200	Pr1	800	100	200	300	200
		Pr2	400	100	100	100	100
LoB2	1,500	Pr3	1,000	500	100	200	200
		Pr4	500	200	100	100	100

Product Dimension must be considered in the calculation

=100/800

	Cu1	Cu2	Cu3	Cu4	Total
Pr1	12.50%	25.00%	37.50%	25.00%	100.00%
Pr2	25.00%	25.00%	25.00%	25.00%	100.00%
Pr3	50.00%	10.00%	20.00%	20.00%	100.00%
Pr4	40.00%	20.00%	20.00%	20.00%	100.00%

=100/500

Driver Calculation is made based on Product

Some More Complex Examples

Case 2 – Allocate on Customer grouping on Product elements

	Alloc.	Filter
Customer	Y	
Product		Elem

				Tier1		Tier2	
				2,675		2,325	
				Cu1	Cu2	Cu3	Cu4
				1,775	900	1,225	1,100
LoB1	2,000	Pr1	1,000	1000*12.5%	1000*25%	1000*37.5%	1000*25%
		Pr2	1,000	1000*25%	1000%25%	1000*25%	1000*25%
LoB2	3,000	Pr3	2,000	2000*50%	2000*10%	2000*20%	2000*20%
		Pr4	1,000	1000*40%	1000*20%	1000*20%	1000*20%

Allocation on Product Driver Calculated Previously

Some More Complex Examples

Case 3 – Allocate on Customer grouping on *Line of Business*

				1,400		1,300	
				Cu1	Cu2	Cu3	Cu4
				900	500	700	600
LoB1	1,200	Pr1	800	100	200	300	200
		Pr2	400	100	100	100	100
LoB2	1,500	Pr3	1,000	500	100	200	200
		Pr4	500	200	100	100	100

Product Dimension must be considered in the calculation

	Alloc.	Filter
Customer	Y	
Product		Node

=100/1200

	Cu1	Cu2	Cu3	Cu4	Total
LOB 1	16.67%	25.00%	33.33%	25.00%	100.00%
LOB 2	46.67%	13.33%	20.00%	20.00%	100.00%

=100/1500

Driver Calculation is made based on Line of Business Aggregation

Some More Complex Examples

Case 3 – Allocate on Customer grouping on *Line of Business*

	Alloc.	Filter
Customer	Y	
Product		Node

				Tier1		Tier2	
				2,633		2,367	
				Cu1	Cu2	Cu3	Cu4
				1,733	900	1,267	1,100
LoB1	2,000	Pr1	1,000	1000*16.7%	1000*25%	1000*33.33%	1000*25%
		Pr2	1,000	1000*16.7%	1000%25%	1000*33.33%	1000*25%
LoB2	3,000	Pr3	2,000	2000*46.67%	2000*13.33%	2000*20%	2000*20%
		Pr4	1,000	1000*46.67%	1000*13.33%	1000*20%	1000*20%

Allocation on Product Driver Calculated Previously

Some More Complex Examples

Case 4 – Allocate on Customer and Product without grouping

				Tier1		Tier2		
				1,400		1,300		
				Cu1	Cu2	Cu3	Cu4	Total
				900	500	700	600	2,700
LoB1	1,200	Pr1	800	100	200	300	200	
		Pr2	400	100	100	100	100	
LoB2	1,500	Pr3	1,000	500	100	200	200	
		Pr4	500	200	100	100	100	
Total			2,700					

=100/2700

	Alloc.	Filter
Customer	Y	
Product	Y	

	Cu1	Cu2	Cu3	Cu4	Total Product
Pr1	3.70%	7.41%	11.11%	7.41%	29.63%
Pr2	3.70%	3.70%	3.70%	3.70%	14.81%
Pr3	18.52%	3.70%	7.41%	7.41%	37.04%
Pr4	7.41%	3.70%	3.70%	3.70%	18.52%
Total Customer	33.33%	18.52%	25.93%	22.22%	100.00%

=100/2700

Allocation Result will be stored considering both the Customer and the Product

Driver Calculation is made on the Total Value by Customer and Product

Some More Complex Examples

Case 4 – Allocate on Customer and Product without grouping

	Alloc.	Filter
Customer	Y	
Product	Y	

				Tier1		Tier2	
				2,593		2,407	
				Cu1	Cu2	Cu3	Cu4
				1,667	926	1,296	1,111
LoB1	2,000	Pr1	1,000	1000*3.70%	1000*7.41%	1000*11.11%	1000*7.41%
		Pr2	1,000	1000*3.70%	1000*3.70%	1000*3.70%	1000*3.70%
LoB2	3,000	Pr3	2,000	2000*18.52%	2000*3.70%	2000*7.41%	2000*7.41%
		Pr4	1,000	1000*7.41%	1000*3.70%	1000*3.70%	1000*3.70%

Allocation on Product/Customer Driver Calculated Previously

Some More Complex Examples

Case 5 – Allocate on Customer and Product, grouping by LOB

				Cu1	Cu2	Cu3	Cu4
				900	500	700	600
LoB1	1,200	Pr1	800	100	200	300	200
		Pr2	400	100	100	100	100
LoB2	1,500	Pr3	1,000	500	100	200	200
		Pr4	500	200	100	100	100

Allocation Result will be stored considering both the Customer and the Product

=100/1200

	Alloc.	Filter
Customer	Y	
Product	Y	Node

	Cu1	Cu2	Cu3	Cu4	Total by LOB
LOB 1	8.33%	16.67%	25.00%	16.67%	100.00%
	8.33%	8.33%	8.33%	8.33%	
LOB 2	33.33%	6.67%	13.33%	13.33%	100.00%
	13.33%	6.67%	6.67%	6.67%	

=100/1500

Driver Calculation will be made considering the Line of Business Grouping

Some More Complex Examples

Case 5 – Allocate on Customer and Product grouping by LOB

	Alloc.	Filter
Customer	Y	
Product	Y	

LoB1	2,000	Pr1	1,000
		Pr2	1,000
LoB2	3,000	Pr3	2,000
		Pr4	1,000

Cu1	Cu2	Cu3	Cu4
1,733	900	1,267	1,100
1000*8.33%	1000*16.67%	1000*25.00%	1000*16.67%
1000*8.33%	1000*8.33%	1000*8.33%	1000*8.33%
2000*33.33%	2000*6.67%	2000*13.33%	2000*13.33%
1000*13.33%	1000*6.67%	1000*6.67%	1000*6.67%

Allocation on Customer/LOB Driver
Calculated Previously

Some More Complex Examples

Recap

Cu1	Cu2	Cu3	Cu4	Total
33.33%	18.52%	25.93%	22.22%	**100%**

Case 1 – Allocate on Customer without grouping

	Cu1	Cu2	Cu3	Cu4	Total
Pr1	12.50%	25.00%	37.50%	25.00%	**100.00%**
Pr2	25.00%	25.00%	25.00%	25.00%	**100.00%**
Pr3	50.00%	10.00%	20.00%	20.00%	**100.00%**
Pr4	40.00%	20.00%	20.00%	20.00%	**100.00%**

Case 2 – Allocate on Customer grouping on Product Elements

Some More Complex Examples

Recap

	Cu1	Cu2	Cu3	Cu4	Total
LOB 1	16.67%	25.00%	33.33%	25.00%	**100.00%**
LOB 2	46.67%	13.33%	20.00%	20.00%	**100.00%**

Case 3 - Allocate on Customer grouping *on Line* of Business

	Cu1	Cu2	Cu3	Cu4	Total Product
Pr1	3.70%	7.41%	11.11%	7.41%	**29.63%**
Pr2	3.70%	3.70%	3.70%	3.70%	**14.81%**
Pr3	18.52%	3.70%	7.41%	7.41%	**37.04%**
Pr4	7.41%	3.70%	3.70%	3.70%	**18.52%**
Total Customer	**33.33%**	**18.52%**	**25.93%**	**22.22%**	**100.00%**

Case 4 - Allocate on Customer and Product, without grouping

Some More Complex Examples

Recap

	Cu1	Cu2	Cu3	Cu4	Total by LOB
LOB 1	8.33%	16.67%	25.00%	16.67%	100.00%
	8.33%	8.33%	8.33%	8.33%	
LOB 2	33.33%	6.67%	13.33%	13.33%	100.00%
	13.33%	6.67%	6.67%	6.67%	

Case 5 – Allocate on Customer and Product, grouping on LOB

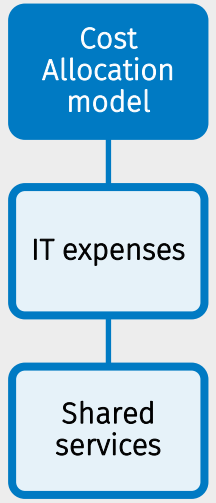
Circular Allocations

Cost Allocation model to fully allocate IT expenses and Shared Services expenses.

Data is allocated on Cost Centers, using a cascading structure:

- IT expenses allocates using the # of closed ticket on the support portal
- Shared services allocates using a manual %

Circular Allocations - Example



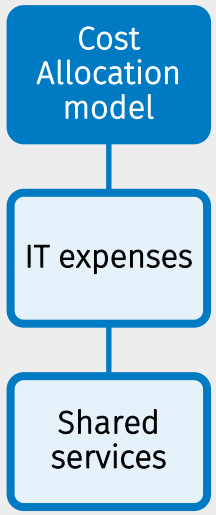
IT allocation

Shared Services allocation

	Expenses	Allocated	Total	Allocated	Total
IT	1000	- 1000	-	500	500
Shared Services	2000	500	2500	-2500	-
Finance	-	300	300	1000	1300
Accounting	-	200	200	1000	1200

At the end of the Second Rules the first Cost Center is not empty!

Circular Allocations



To Solve this issue, we can define the allocation as circular

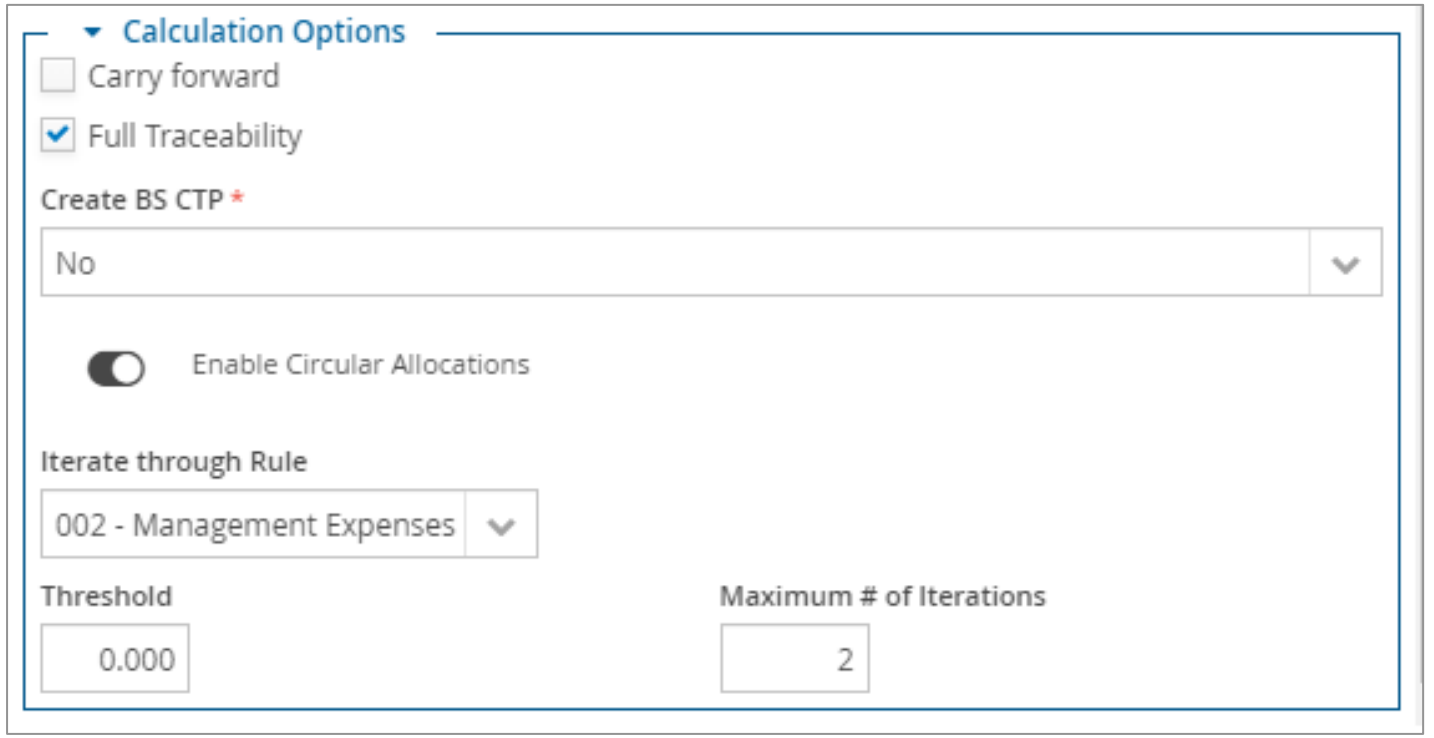
The allocation will run for a desired number of times until the receiver is zeroed out

Circular Allocations - Settings

Defined at Rule level

Works ONLY with cascading rules

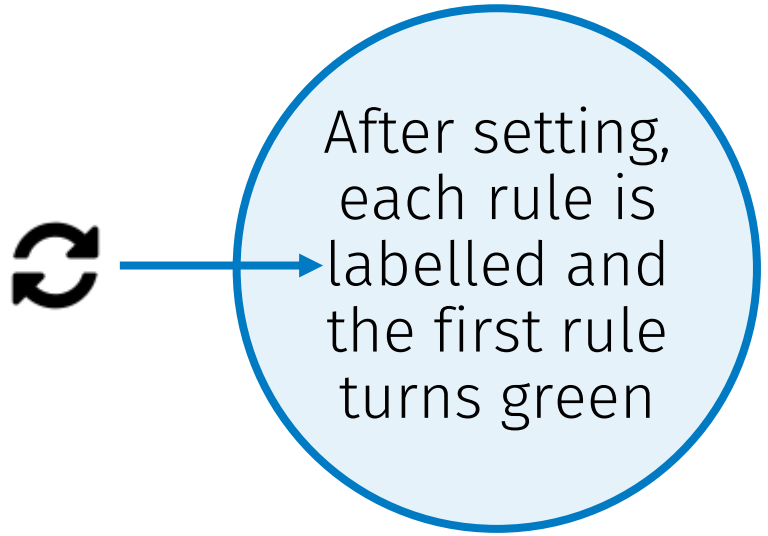
Circular Allocations – Settings: Parameters

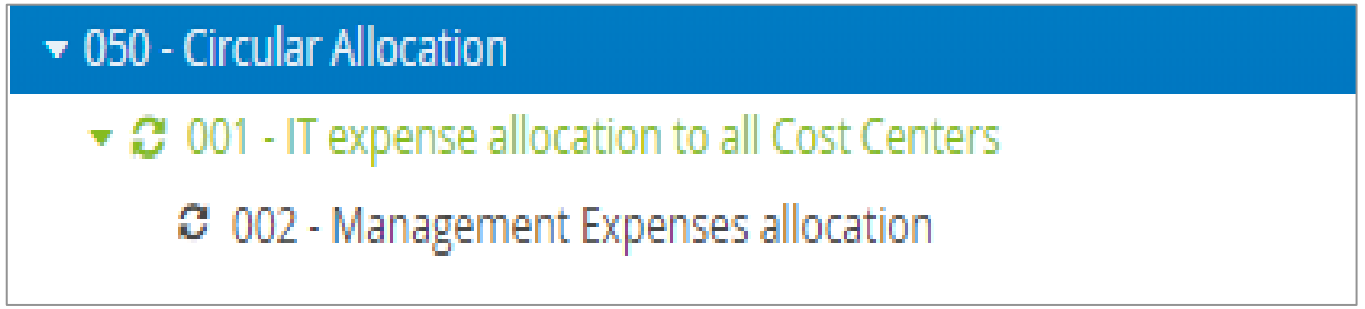


3 parameters

- Iterate through rule
- Threshold
- Maximum # of iterations

Circular Allocations – Settings: after





Now it’s your turn

Exercise # 8





Takeaways

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- 2 tickets CCH Tagetik inTouch
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Connect member contribution

- Customer quote
- 4 or more reference calls
- 2 analyst reviews
- Feedback loop

Click Here

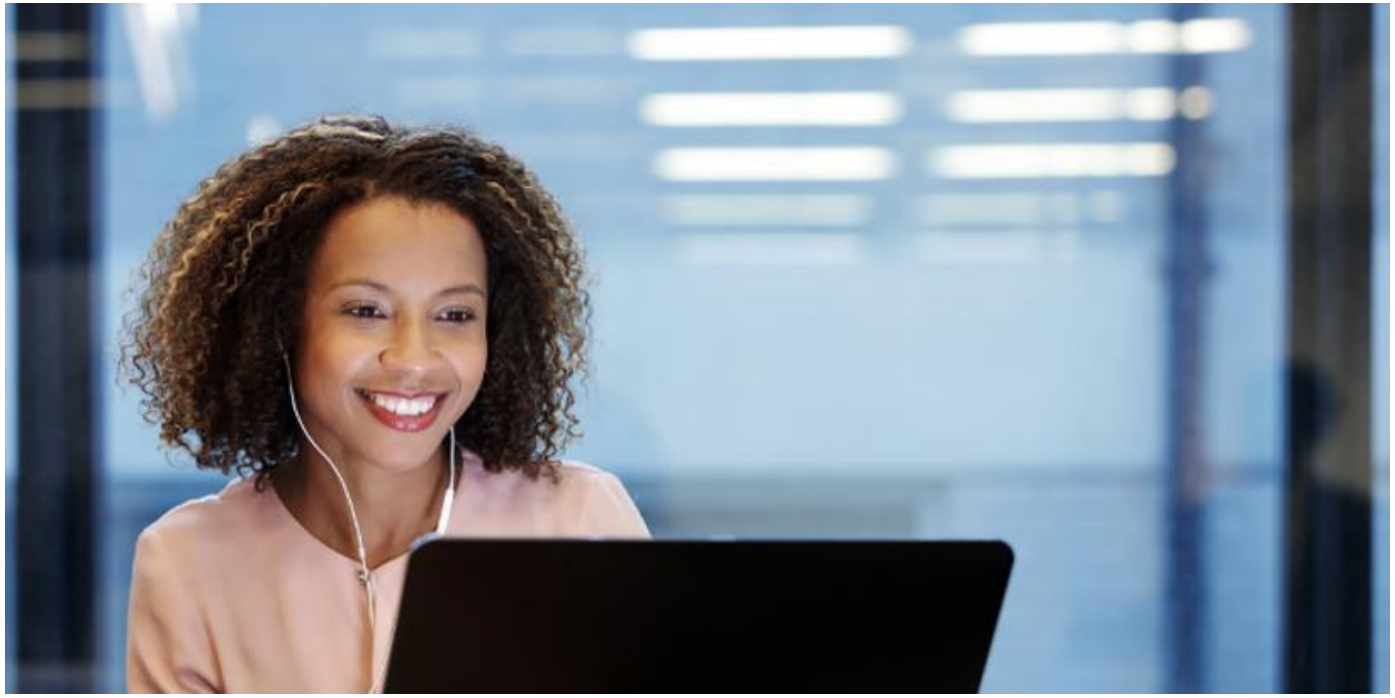
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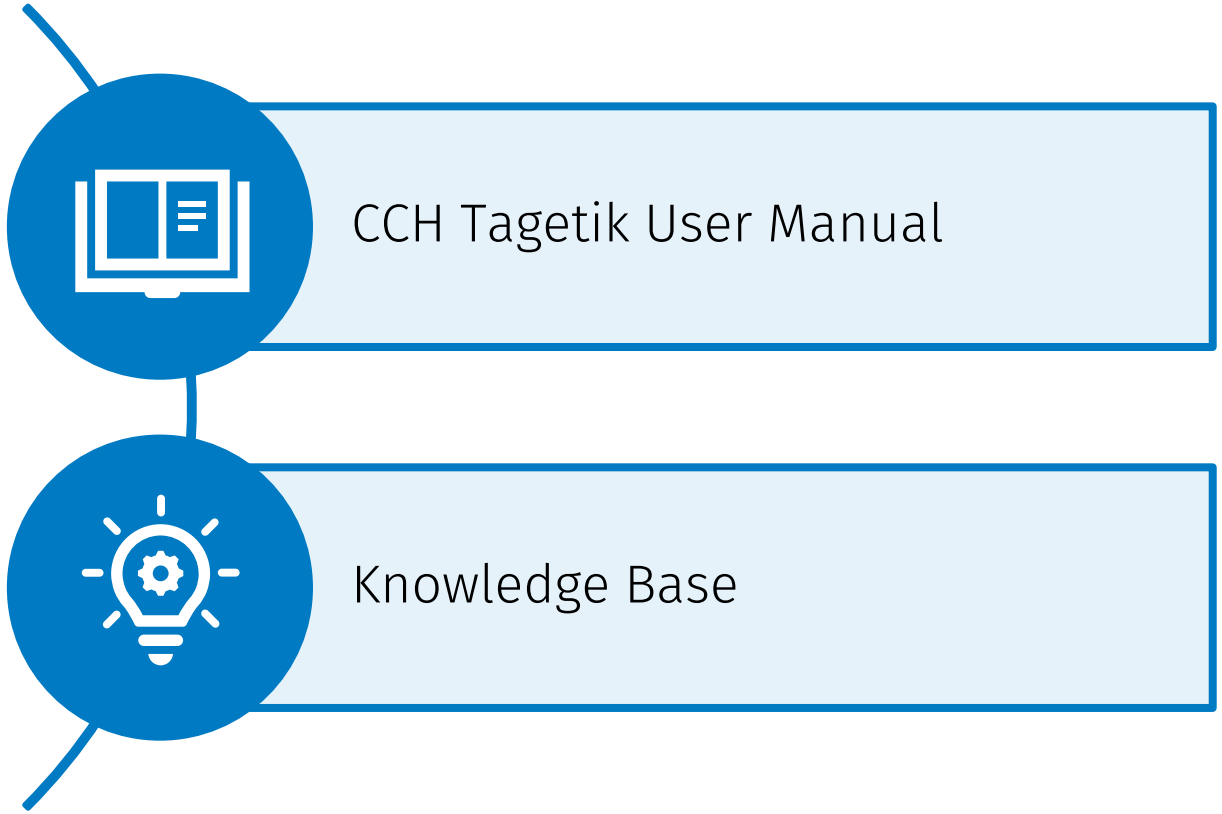


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